Lampiran A

Perhitungan Matriks Densitas Tereduksi

A.1 Matriks densitas tereduksi dua qubit

Bentuk umum masing-masing matriks densitas tereduksi telah didapat pada persamaan (2.83) dan (2.84)

$$
\begin{aligned}
\rho_A &= \sum_{m=0}^{1} \sum_{i_1,i_2=0}^{1} \alpha_{i_1 m} \alpha^*_{i_2 m} \left| i_1 \right\rangle \left\langle i_2 \right| \\
\rho_B &= \sum_{m=0}^{1} \sum_{j_1,j_2=0}^{1} \alpha_{m j_1} \alpha^*_{m j_2} \left| j_1 \right\rangle \left\langle j_2 \right| .
\end{aligned}
$$

A.1.1 Keadaan Terpisah

Untuk keadaan terpisah, substitusi nilai koefisien pada persamaan (4.36) pada masing-masing matriks densitas tereduksi ρ_A dan ρ_B

$$\alpha_{ij} = a_i b_j .$$

Matriks densitas tereduksi $\rho_A^{(sep)}$

$$
\begin{aligned}
\rho_A^{(sep)} &= \sum_{m=0}^{1} \sum_{i_1,i_2=0}^{1} a_{i_1} b_m a^*_{i_2} b^*_m \left| i_1 \right\rangle \left\langle i_2 \right| \\
&= \begin{pmatrix} a_0 b_0 a_0^* b_0^* + a_0 b_1 a_0^* b_1^* & a_0 b_0 a_1^* b_0^* + a_0 b_1 a_1^* b_1^* \\ a_1 b_0 a_0^* b_0^* + a_1 b_1 a_0^* b_1^* & a_1 b_0 a_1^* b_0^* + a_1 b_1 a_1^* b_1^* \end{pmatrix} \\
&= \begin{pmatrix} \left|a_0\right|^2 \left(\left|b_0\right|^2 + \left|b_1\right|^2\right) & a_0 a_1^* \left(\left|b_0\right|^2 + \left|b_1\right|^2\right) \\ a_1 a_0^* \left(\left|b_0\right|^2 + \left|b_1\right|^2\right) & \left|a_1\right|^2 \left(\left|b_0\right|^2 + \left|b_1\right|^2\right) \end{pmatrix} \\
\rho_A^{(sep)} &= \begin{pmatrix} \left|a_0\right|^2 & a_0 a_1^* \\ a_1 a_0^* & \left|a_1\right|^2 \end{pmatrix} . \qquad \text{(A.1)}
\end{aligned}
$$

Matriks densitas tereduksi $\rho_B^{(sep)}$

$$\begin{aligned}
\rho_B^{(sep)} &= \sum_{m=0}^{1} \sum_{j_1,j_2=0}^{1} a_m b_{j_1} a_m^* b_{j_2}^* \left|j_1\right\rangle \left\langle j_2\right| \\
&= \begin{pmatrix} a_0 b_0 a_0^* b_0^* + a_1 b_0 a_1^* b_0^* & a_0 b_0 a_0^* b_1^* + a_1 b_0 a_1^* b_1^* \\ a_0 b_1 a_0^* b_0^* + a_1 b_1 a_1^* b_0^* & a_0 b_1 a_0^* b_1^* + a_1 b_1 a_1^* b_1^* \end{pmatrix} \\
&= \begin{pmatrix} \left(|a_0|^2 + |a_1|^2\right) |b_0|^2 & \left(|a_0|^2 + |a_1|^2\right) b_0 b_1^* \\ \left(|a_0|^2 + |a_1|^2\right) b_1 b_0^* & \left(|a_0|^2 + |a_1|^2\right) |b_1|^2 \end{pmatrix} \\
\rho_B^{(sep)} &= \begin{pmatrix} |b_0|^2 & b_0 b_1^* \\ b_1 b_0^* & |b_1|^2 \end{pmatrix}.
\end{aligned} \tag{A.2}$$

A.1.2 Keadaan Terbelit

Untuk keadaan terbelit, nilai koefisien tidak bisa dinyatakan dalam persamaan (2.93)

$$\alpha_{ij} \neq a_i b_j.$$

Matriks densitas tereduksi $\rho_A^{(ent)}$

$$\begin{aligned}
\rho_A^{(ent)} &= \sum_{m=0}^{1} \sum_{i_1,i_2=0}^{1} \alpha_{i_1 m} \alpha_{i_2 m}^* \left|i_1\right\rangle \left\langle i_2\right| \\
&= \begin{pmatrix} |\alpha_{00}|^2 + |\alpha_{01}|^2 & \alpha_{00}\alpha_{10}^* + \alpha_{01}\alpha_{11}^* \\ \alpha_{10}\alpha_{00}^* + \alpha_{11}\alpha_{01}^* & |\alpha_{10}|^2 + |\alpha_{11}|^2 \end{pmatrix}.
\end{aligned} \tag{A.3}$$

Matriks densitas tereduksi $\rho_B^{(ent)}$

$$\begin{aligned}
\rho_B^{(ent)} &= \sum_{m=0}^{1} \sum_{j_1,j_2=0}^{1} a_m b_{j_1} a_m^* b_{j_2}^* \left|j_1\right\rangle \left\langle j_2\right| \\
&= \begin{pmatrix} |\alpha_{00}|^2 + |\alpha_{10}|^2 & \alpha_{00}\alpha_{01}^* + \alpha_{10}\alpha_{11}^* \\ \alpha_{01}\alpha_{00}^* + \alpha_{11}\alpha_{10}^* & |\alpha_{01}|^2 + |\alpha_{11}|^2 \end{pmatrix}.
\end{aligned} \tag{A.4}$$

A.2 Matriks densitas tereduksi tiga qubit

Bentuk umum masing-masing matriks densitas tereduksi telah didapat pada persamaan (2.85) sampai persamaan (2.90)

$$\begin{aligned}
\rho_A &= \sum_{m,n=0}^{1} \sum_{i_1,i_2=0}^{1} \alpha_{i_1 mn} \alpha_{i_2 mn}^* \left|i_1\right\rangle \left\langle i_2\right| \\
\rho_B &= \sum_{m,n=0}^{1} \sum_{j_1,j_2=0}^{1} \alpha_{m j_1 n} \alpha_{m j_2 n}^* \left|j_1\right\rangle \left\langle j_2\right| \\
\rho_C &= \sum_{m,n=0}^{1} \sum_{k_1,k_2=0}^{1} \alpha_{mn k_1} \alpha_{mn k_2}^* \left|k_1\right\rangle \left\langle k_2\right|
\end{aligned}$$

$$\rho_{AB} = \sum_{m=0}^{1} \sum_{i_1,j_1=0}^{1} \sum_{i_2,j_2=0}^{1} \alpha_{i_1 j_1 m} \alpha^*_{i_2 j_2 m} \left| i_1 j_1 \right\rangle \left\langle i_2 j_2 \right|$$

$$\rho_{AC} = \sum_{m=0}^{1} \sum_{i_1,k_1=0}^{1} \sum_{i_2,k_2=0}^{1} \alpha_{i_1 m k_1} \alpha^*_{i_2 m k_2} \left| i_1 k_1 \right\rangle \left\langle i_2 k_2 \right|$$

$$\rho_{BC} = \sum_{m=0}^{1} \sum_{j_1,k_1=0}^{1} \sum_{j_2,k_2=0}^{1} \alpha_{m j_1 k_1} \alpha^*_{m j_2 k_2} \left| j_1 k_1 \right\rangle \left\langle j_2 k_2 \right| .$$

A.2.1 Keadaan Terpisah Sepenuhnya

Untuk keadaan terpisah sepenuhnya, substitusi nilai koefisien persamaan (4.39) pada masing-masing matriks densitas tereduksi $\rho_A, \rho_B . \rho_C, \rho_{AB}, \rho_{AC}$ dan ρ_{BC}

$$\alpha_{ijk} = a_i b_j c_k.$$

Matriks densitas tereduksi $\rho_A^{(sep)}$

$$\begin{aligned}
\rho_A^{(sep)} &= \sum_{m,n=0}^{1} \sum_{i_1,i_2=0}^{1} a_{i_1} b_m c_n a_{i_2}^* b_m^* c_n^* \left|i_1\right\rangle \left\langle i_2\right| \\
&= \begin{pmatrix} a_0b_0c_0a_0^*b_0^*c_0^* + a_0b_0c_1a_0^*b_0^*c_1^* + a_0b_1c_0a_0^*b_1^*c_0^* + a_0b_1c_1a_0^*b_1^*c_1^* & a_0b_0c_0a_1^*b_0^*c_0^* + a_0b_0c_1a_1^*b_0^*c_1^* + a_0b_1c_0a_1^*b_1^*c_0^* + a_0b_1c_1a_1^*b_1^*c_1^* \\ a_1b_0c_0a_0^*b_0^*c_0^* + a_1b_0c_1a_0^*b_0^*c_1^* + a_1b_1c_0a_0^*b_1^*c_0^* + a_1b_1c_1a_0^*b_1^*c_1^* & a_1b_0c_0a_1^*b_0^*c_0^* + a_1b_0c_1a_1^*b_0^*c_1^* + a_1b_1c_0a_1^*b_1^*c_0^* + a_1b_1c_1a_1^*b_1^*c_1^* \end{pmatrix} \\
&= \begin{pmatrix} |a_0|^2 \left(|b_0|^2 + |b_1|^2\right)\left(|c_0|^2 + |c_1|^2\right) & a_0a_1^* \left(|b_0|^2 + |b_1|^2\right)\left(|c_0|^2 + |c_1|^2\right) \\ a_1a_0^* \left(|b_0|^2 + |b_1|^2\right)\left(|c_0|^2 + |c_1|^2\right) & |a_1|^2 \left(|b_0|^2 + |b_1|^2\right)\left(|c_0|^2 + |c_1|^2\right) \end{pmatrix} \\
\rho_A^{(sep)} &= \begin{pmatrix} |a_0|^2 & a_0a_1^* \\ a_1a_0^* & |a_1|^2 \end{pmatrix}.
\end{aligned} \tag{A.5}$$

Matriks densitas tereduksi $\rho_B^{(sep)}$

$$\begin{aligned}
\rho_B^{(sep)} &= \sum_{m,n=0}^{1} \sum_{j_1,j_2=0}^{1} a_m b_{j_1} c_n a_m^* b_{j_2}^* c_n^* \left|j_1\right\rangle \left\langle j_2\right| \\
&= \begin{pmatrix} a_0b_0c_0a_0^*b_0^*c_0^* + a_0b_0c_1a_0^*b_0^*c_1^* + a_1b_0c_0a_1^*b_0^*c_0^* + a_1b_0c_1a_1^*b_0^*c_1^* & a_0b_0c_0a_0^*b_1^*c_0^* + a_0b_0c_1a_0^*b_1^*c_1^* + a_1b_0c_0a_1^*b_1^*c_0^* + a_1b_0c_1a_1^*b_1^*c_1^* \\ a_0b_1c_0a_0^*b_0^*c_0^* + a_0b_1c_1a_0^*b_0^*c_1^* + a_1b_1c_0a_1^*b_0^*c_0^* + a_1b_1c_1a_1^*b_0^*c_1^* & a_0b_1c_0a_0^*b_1^*c_0^* + a_0b_1c_1a_0^*b_1^*c_1^* + a_1b_1c_0a_1^*b_1^*c_0^* + a_1b_1c_1a_1^*b_1^*c_1^* \end{pmatrix} \\
&= \begin{pmatrix} \left(|a_0|^2 + |a_1|^2\right)|b_0|^2\left(|c_0|^2 + |c_1|^2\right) & \left(|a_0|^2 + |a_1|^2\right) b_0b_1^* \left(|c_0|^2 + |c_1|^2\right) \\ \left(|a_0|^2 + |a_1|^2\right) b_1b_0^* \left(|c_0|^2 + |c_1|^2\right) & \left(|a_0|^2 + |a_1|^2\right)|b_1|^2\left(|c_0|^2 + |c_1|^2\right) \end{pmatrix} \\
\rho_B^{(sep)} &= \begin{pmatrix} |b_0|^2 & b_0b_1^* \\ b_1b_0^* & |b_1|^2 \end{pmatrix}.
\end{aligned} \tag{A.6}$$

Matriks densitas tereduksi $\rho_C^{(sep)}$

$$
\begin{aligned}
\rho_C^{(sep)} &= \sum_{m,n=0}^{1} \sum_{k_1,k_2=0}^{1} a_m b_n c_{k_1} a_m^* b_n^* c_{k_2}^* \left|k_1\right\rangle \left\langle k_2\right| \\
&= \begin{pmatrix} a_0b_0c_0a_0^*b_0^*c_0^* + a_0b_1c_0a_0^*b_1^*c_0^* + a_1b_0c_0a_1^*b_0^*c_0^* + a_1b_1c_0a_1^*b_1^*c_0^* & a_0b_0c_0a_0^*b_0^*c_1^* + a_0b_1c_0a_0^*b_1^*c_1^* + a_1b_0c_0a_1^*b_0^*c_1^* + a_1b_1c_0a_1^*b_1^*c_1^* \\ a_0b_0c_1a_0^*b_0^*c_0^* + a_0b_1c_1a_0^*b_1^*c_0^* + a_1b_0c_1a_1^*b_0^*c_0^* + a_1b_1c_1a_1^*b_1^*c_0^* & a_0b_0c_1a_0^*b_0^*c_1^* + a_0b_1c_1a_0^*b_1^*c_1^* + a_1b_0c_1a_1^*b_0^*c_1^* + a_1b_1c_1a_1^*b_1^*c_1^* \end{pmatrix} \\
&= \begin{pmatrix} \left(|a_0|^2 + |a_1|^2\right)\left(|b_0|^2 + |b_1|^2\right)|c_0|^2 & \left(|a_0|^2 + |a_1|^2\right)\left(|b_0|^2 + |b_1|^2\right)c_0c_1^* \\ \left(|a_0|^2 + |a_1|^2\right)\left(|b_0|^2 + |b_1|^2\right)c_1c_0^* & \left(|a_0|^2 + |a_1|^2\right)\left(|b_0|^2 + |b_1|^2\right)|c_1|^2 \end{pmatrix} \\
\rho_C^{(sep)} &= \begin{pmatrix} |c_0|^2 & c_0c_1^* \\ c_1c_0^* & |c_1|^2 \end{pmatrix}.
\end{aligned} \tag{A.7}
$$

Matriks densitas tereduksi $\rho_{AB}^{(sep)}$

$$
\begin{aligned}
\rho_{AB}^{(sep)} &= \sum_{m=0}^{1} \sum_{i_1,i_2=0}^{1} \sum_{j_1,j_2=0}^{1} a_{i_1} b_{j_1} c_m a_{i_2}^* b_{j_2}^* c_m^* \left|i_1 j_1\right\rangle \left\langle i_2 j_2\right| \\
&= \begin{pmatrix}
a_0b_0c_0a_0^*b_0^*c_0^* + a_0b_0c_1a_0^*b_0^*c_1^* & a_0b_0c_0a_0^*b_1^*c_0^* + a_0b_0c_1a_0^*b_1^*c_1^* & a_0b_0c_0a_1^*b_0^*c_0^* + a_0b_0c_1a_1^*b_0^*c_1^* & a_0b_0c_0a_1^*b_1^*c_0^* + a_0b_0c_1a_1^*b_1^*c_1^* \\
a_0b_1c_0a_0^*b_0^*c_0^* + a_0b_1c_1a_0^*b_0^*c_1^* & a_0b_1c_0a_0^*b_1^*c_0^* + a_0b_1c_1a_0^*b_1^*c_1^* & a_0b_1c_0a_1^*b_0^*c_0^* + a_0b_1c_1a_1^*b_0^*c_1^* & a_0b_1c_0a_1^*b_1^*c_0^* + a_0b_1c_1a_1^*b_1^*c_1^* \\
a_1b_0c_0a_0^*b_0^*c_0^* + a_1b_0c_1a_0^*b_0^*c_1^* & a_1b_0c_0a_0^*b_1^*c_0^* + a_1b_0c_1a_0^*b_1^*c_1^* & a_1b_0c_0a_1^*b_0^*c_0^* + a_1b_0c_1a_1^*b_0^*c_1^* & a_1b_0c_0a_1^*b_1^*c_0^* + a_1b_0c_1a_1^*b_1^*c_1^* \\
a_1b_1c_0a_0^*b_0^*c_0^* + a_1b_1c_1a_0^*b_0^*c_1^* & a_1b_1c_0a_0^*b_1^*c_0^* + a_1b_1c_1a_0^*b_1^*c_1^* & a_1b_1c_0a_1^*b_0^*c_0^* + a_1b_1c_1a_1^*b_0^*c_1^* & a_1b_1c_0a_1^*b_1^*c_0^* + a_1b_1c_1a_1^*b_1^*c_1^*
\end{pmatrix} \\
&= \begin{pmatrix}
|a_0|^2|b_0|^2\left(|c_0|^2 + |c_1|^2\right) & |a_0|^2b_0b_1^*\left(|c_0|^2 + |c_1|^2\right) & a_0a_1^*|b_0|^2\left(|c_0|^2 + |c_1|^2\right) & a_0a_1^*b_0b_1^*\left(|c_0|^2 + |c_1|^2\right) \\
|a_0|^2b_1b_0^*\left(|c_0|^2 + |c_1|^2\right) & |a_0|^2|b_1|^2\left(|c_0|^2 + |c_1|^2\right) & a_0a_1^*b_1b_0^*\left(|c_0|^2 + |c_1|^2\right) & a_0a_1^*|b_1|^2\left(|c_0|^2 + |c_1|^2\right) \\
a_1a_0^*|b_0|^2\left(|c_0|^2 + |c_1|^2\right) & a_1a_0^*b_0b_1^*\left(|c_0|^2 + |c_1|^2\right) & |a_1|^2|b_0|^2\left(|c_0|^2 + |c_1|^2\right) & |a_1|^2b_0b_1^*\left(|c_0|^2 + |c_1|^2\right) \\
a_1a_0^*b_1b_0^*\left(|c_0|^2 + |c_1|^2\right) & a_1a_0^*|b_1|^2\left(|c_0|^2 + |c_1|^2\right) & |a_1|^2b_1b_0^*\left(|c_0|^2 + |c_1|^2\right) & |a_1|^2|b_1|^2\left(|c_0|^2 + |c_1|^2\right)
\end{pmatrix}
\end{aligned}
$$

$$\rho_{AB}^{(sep)} = \begin{pmatrix} |a_0|^2|b_0|^2 & |a_0|^2 b_0 b_1^* & a_0 a_1^* |b_0|^2 & a_0 a_1^* b_0 b_1^* \\ |a_0|^2 b_1 b_0^* & |a_0|^2 |b_1|^2 & a_0 a_1^* b_1 b_0^* & a_0 a_1^* |b_1|^2 \\ a_1 a_0^* |b_0|^2 & a_1 a_0^* b_0 b_1^* & |a_1|^2 |b_0|^2 & |a_1|^2 b_0 b_1^* \\ a_1 a_0^* b_1 b_0^* & a_1 a_0^* |b_1|^2 & |a_1|^2 b_1 b_0^* & |a_1|^2 |b_1|^2 \end{pmatrix}. \tag{A.8}$$

Matriks densitas tereduksi $\rho_{AC}^{(sep)}$

$$\begin{aligned}
\rho_{AC}^{(sep)} &= \sum_{m=0}^{1} \sum_{i_1,i_2=0}^{1} \sum_{k_1,k_2=0}^{1} a_{i_1} b_m c_{k_1} a_{i_2}^* b_m^* c_{k_2}^* \left| i_1 k_1 \right\rangle \left\langle i_2 k_2 \right| \\
&= \begin{pmatrix} a_0b_0c_0a_0^*b_0^*c_0^* + a_0b_1c_0a_0^*b_1^*c_0^* & a_0b_0c_0a_0^*b_0^*c_1^* + a_0b_1c_0a_0^*b_1^*c_1^* & a_0b_0c_0a_1^*b_0^*c_0^* + a_0b_1c_0a_1^*b_1^*c_0^* & a_0b_0c_0a_1^*b_0^*c_1^* + a_0b_1c_0a_1^*b_1^*c_1^* \\ a_0b_0c_1a_0^*b_0^*c_0^* + a_0b_1c_1a_0^*b_1^*c_0^* & a_0b_0c_1a_0^*b_0^*c_1^* + a_0b_1c_1a_0^*b_1^*c_1^* & a_0b_0c_1a_1^*b_0^*c_0^* + a_0b_1c_1a_1^*b_1^*c_0^* & a_0b_0c_1a_1^*b_0^*c_1^* + a_0b_1c_1a_1^*b_1^*c_1^* \\ a_1b_0c_0a_0^*b_0^*c_0^* + a_1b_1c_0a_0^*b_1^*c_0^* & a_1b_0c_0a_0^*b_0^*c_1^* + a_1b_1c_0a_0^*b_1^*c_1^* & a_1b_0c_0a_1^*b_0^*c_0^* + a_1b_1c_0a_1^*b_1^*c_0^* & a_1b_0c_0a_1^*b_0^*c_1^* + a_1b_1c_0a_1^*b_1^*c_1^* \\ a_1b_0c_1a_0^*b_0^*c_0^* + a_1b_1c_1a_0^*b_1^*c_0^* & a_1b_0c_1a_0^*b_0^*c_1^* + a_1b_1c_1a_0^*b_1^*c_1^* & a_1b_0c_1a_1^*b_0^*c_0^* + a_1b_1c_1a_1^*b_1^*c_0^* & a_1b_0c_1a_1^*b_0^*c_1^* + a_1b_1c_1a_1^*b_1^*c_1^* \end{pmatrix} \\
&= \begin{pmatrix} |a_0|^2 \left(|b_0|^2 + |b_1|^2\right) |c_0|^2 & |a_0|^2 \left(|b_0|^2 + |b_1|^2\right) c_0 c_1^* & a_0 a_1^* \left(|b_0|^2 + |b_1|^2\right) |c_0|^2 & a_0 a_1^* \left(|b_0|^2 + |b_1|^2\right) c_0 c_1^* \\ |a_0|^2 \left(|b_0|^2 + |b_1|^2\right) c_1 c_0^* & |a_0|^2 \left(|b_0|^2 + |b_1|^2\right) |c_1|^2 & a_0 a_1^* \left(|b_0|^2 + |b_1|^2\right) c_1 c_0^* & a_0 a_1^* \left(|b_0|^2 + |b_1|^2\right) |c_1|^2 \\ a_1 a_0^* \left(|b_0|^2 + |b_1|^2\right) |c_0|^2 & a_1 a_0^* \left(|b_0|^2 + |b_1|^2\right) c_0 c_1^* & |a_1|^2 \left(|b_0|^2 + |b_1|^2\right) |c_0|^2 & |a_1|^2 \left(|b_0|^2 + |b_1|^2\right) c_0 c_1^* \\ a_1 a_0^* \left(|b_0|^2 + |b_1|^2\right) c_1 c_0^* & a_1 a_0^* \left(|b_0|^2 + |b_1|^2\right) |c_1|^2 & |a_1|^2 \left(|b_0|^2 + |b_1|^2\right) c_1 c_0^* & |a_1|^2 \left(|b_0|^2 + |b_1|^2\right) |c_1|^2 \end{pmatrix}
\end{aligned}$$

$$\rho_{AC}^{(sep)} = \begin{pmatrix} |a_0|^2|c_0|^2 & |a_0|^2 c_0 c_1^* & a_0 a_1^* |c_0|^2 & a_0 a_1^* c_0 c_1^* \\ |a_0|^2 c_1 c_0^* & |a_0|^2 |c_1|^2 & a_0 a_1^* c_1 c_0^* & a_0 a_1^* |c_1|^2 \\ a_1 a_0^* |c_0|^2 & a_1 a_0^* c_0 c_1^* & |a_1|^2 |c_0|^2 & |a_1|^2 c_0 c_1^* \\ a_1 a_0^* c_1 c_0^* & a_1 a_0^* |c_1|^2 & |a_1|^2 c_1 c_0^* & |a_1|^2 |c_1|^2 \end{pmatrix}. \tag{A.9}$$

Matriks densitas tereduksi $\rho_{BC}^{(sep)}$ dapat ditulis menjadi

$$\rho_{BC}^{(sep)} = \sum_{m=0}^{1} \sum_{j_1,j_2=0}^{1} \sum_{k_1,k_2=0}^{1} a_m b_{j_1} c_{k_1} a_m^* b_{j_2}^* c_{k_2}^* \left| j_1 k_1 \right\rangle \left\langle j_2 k_2 \right|$$

$$= \begin{pmatrix} a_0b_0c_0a_0^*b_0^*c_0^* + a_1b_0c_0a_1^*b_0^*c_0^* & a_0b_0c_0a_0^*b_0^*c_1^* + a_1b_0c_0a_1^*b_0^*c_1^* & a_0b_0c_0a_0^*b_1^*c_0^* + a_1b_0c_0a_1^*b_1^*c_0^* & a_0b_0c_0a_0^*b_1^*c_1^* + a_1b_0c_0a_1^*b_1^*c_1^* \\ a_0b_0c_1a_0^*b_0^*c_0^* + a_1b_0c_1a_1^*b_0^*c_0^* & a_0b_0c_1a_0^*b_0^*c_1^* + a_1b_0c_1a_1^*b_0^*c_1^* & a_0b_0c_1a_0^*b_1^*c_0^* + a_1b_0c_1a_1^*b_1^*c_0^* & a_0b_0c_1a_0^*b_1^*c_1^* + a_1b_0c_1a_1^*b_1^*c_1^* \\ a_0b_1c_0a_0^*b_0^*c_0^* + a_1b_1c_0a_1^*b_0^*c_0^* & a_0b_1c_0a_0^*b_0^*c_1^* + a_1b_1c_0a_1^*b_0^*c_1^* & a_0b_1c_0a_0^*b_1^*c_0^* + a_1b_1c_0a_1^*b_1^*c_0^* & a_0b_1c_0a_0^*b_1^*c_1^* + a_1b_1c_0a_1^*b_1^*c_1^* \\ a_0b_1c_1a_0^*b_0^*c_0^* + a_1b_1c_1a_1^*b_0^*c_0^* & a_0b_1c_1a_0^*b_0^*c_1^* + a_1b_1c_1a_1^*b_0^*c_1^* & a_0b_1c_1a_0^*b_1^*c_0^* + a_1b_1c_1a_1^*b_1^*c_0^* & a_0b_1c_1a_0^*b_1^*c_1^* + a_1b_1c_1a_1^*b_1^*c_1^* \end{pmatrix}$$

$$= \begin{pmatrix} \left(|a_0|^2 + |a_1|^2\right)|b_0|^2|c_0|^2 & \left(|a_0|^2 + |a_1|^2\right)|b_0|^2c_0c_1^* & \left(|a_0|^2 + |a_1|^2\right)b_0b_1^*|c_0|^2 & \left(|a_0|^2 + |a_1|^2\right)b_0b_1^*c_0c_1^* \\ \left(|a_0|^2 + |a_1|^2\right)|b_0|^2c_1c_0^* & \left(|a_0|^2 + |a_1|^2\right)|b_0|^2|c_1|^2 & \left(|a_0|^2 + |a_1|^2\right)b_0b_1^*c_1c_0^* & \left(|a_0|^2 + |a_1|^2\right)b_0b_1^*|c_1|^2 \\ \left(|a_0|^2 + |a_1|^2\right)b_1b_0^*|c_0|^2 & \left(|a_0|^2 + |a_1|^2\right)b_1b_0^*c_0c_1^* & \left(|a_0|^2 + |a_1|^2\right)|b_1|^2|c_0|^2 & \left(|a_0|^2 + |a_1|^2\right)|b_1|^2c_0c_1^* \\ \left(|a_0|^2 + |a_1|^2\right)b_1b_0^*c_1c_0^* & \left(|a_0|^2 + |a_1|^2\right)b_1b_0^*|c_1|^2 & \left(|a_0|^2 + |a_1|^2\right)|b_1|^2c_1c_0^* & \left(|a_0|^2 + |a_1|^2\right)|b_1|^2|c_1|^2 \end{pmatrix}$$

$$\rho_{BC}^{(sep)} = \begin{pmatrix} |b_0|^2|c_0|^2 & |b_0|^2c_0c_1^* & b_0b_1^*|c_0|^2 & b_0b_1^*c_0c_1^* \\ |b_0|^2c_1c_0^* & |b_0|^2|c_1|^2 & b_0b_1^*c_1c_0^* & b_0b_1^*|c_1|^2 \\ b_1b_0^*|c_0|^2 & b_1b_0^*c_0c_1^* & |b_1|^2|c_0|^2 & |b_1|^2c_0c_1^* \\ b_1b_0^*c_1c_0^* & b_1b_0^*|c_1|^2 & |b_1|^2c_1c_0^* & |b_1|^2|c_1|^2 \end{pmatrix}. \tag{A.10}$$

A.2.2 Keadaan Terbelit Sepenuhnya

Untuk keadaan terbelit sepenuhnya, nilai koefisien tidak bisa dinyatakan seperti persamaan (4.22)

$$\alpha_{ijk} \neq a_ib_jc_k.$$

Matriks densitas tereduksi $\rho_A^{(ent)}$

$$\rho_A^{(ent)} = \sum_{m,n=0}^{1} \sum_{i_1,i_2=0}^{1} \alpha_{i_1mn}\alpha_{i_2mn}^* |i_1\rangle\langle i_2|$$

$$= \begin{pmatrix} |\alpha_{000}|^2 + |\alpha_{001}|^2 + |\alpha_{010}|^2 + |\alpha_{011}|^2 & \alpha_{000}\alpha_{100}^* + \alpha_{001}\alpha_{101}^* + \alpha_{010}\alpha_{110}^* + \alpha_{011}\alpha_{111}^* \\ \alpha_{100}\alpha_{000}^* + \alpha_{101}\alpha_{001}^* + \alpha_{110}\alpha_{010}^* + \alpha_{111}\alpha_{011}^* & |\alpha_{100}|^2 + |\alpha_{101}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2 \end{pmatrix}. \tag{A.11}$$

Matriks densitas tereduksi $\rho_B^{(ent)}$

$$\rho_B^{(ent)} = \sum_{m,n=0}^{1} \sum_{j_1,j_2=0}^{1} \alpha_{mj_1n}\alpha_{mj_2n}^* |j_1\rangle\langle j_2|$$

$$
= \begin{pmatrix} |\alpha_{000}|^2 + |\alpha_{001}|^2 + |\alpha_{100}|^2 + |\alpha_{101}|^2 & \alpha_{000}\alpha_{010}^* + \alpha_{001}\alpha_{011}^* + \alpha_{100}\alpha_{110}^* + \alpha_{101}\alpha_{111}^* \\ \alpha_{010}\alpha_{000}^* + \alpha_{011}\alpha_{001}^* + \alpha_{110}\alpha_{100}^* + \alpha_{111}\alpha_{101}^* & |\alpha_{010}|^2 + |\alpha_{011}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2 \end{pmatrix}. \tag{A.12}
$$

Matriks densitas tereduksi $\rho_C^{(ent)}$

$$
\begin{aligned}
\rho_C^{(ent)} &= \sum_{m,n=0}^{1} \sum_{k_1,k_2=0}^{1} \alpha_{mnk_1}\alpha_{mnk_2}^* \left|k_1\right\rangle \left\langle k_2\right| \\
&= \begin{pmatrix} |\alpha_{000}|^2 + |\alpha_{010}|^2 + |\alpha_{100}|^2 + |\alpha_{110}|^2 & \alpha_{000}\alpha_{001}^* + \alpha_{010}\alpha_{011}^* + \alpha_{100}\alpha_{101}^* + \alpha_{110}\alpha_{111}^* \\ \alpha_{001}\alpha_{000}^* + \alpha_{011}\alpha_{010}^* + \alpha_{101}\alpha_{100}^* + \alpha_{111}\alpha_{110}^* & |\alpha_{001}|^2 + |\alpha_{011}|^2 + |\alpha_{101}|^2 + |\alpha_{111}|^2 \end{pmatrix}.
\end{aligned} \tag{A.13}
$$

Matriks densitas tereduksi $\rho_{AB}^{(ent)}$

$$
\begin{aligned}
\rho_{AB}^{(ent)} &= \sum_{m=0}^{1} \sum_{i_1,j_1=0}^{1} \sum_{i_2,j_2=0}^{1} \alpha_{i_1j_1m}\alpha_{i_2j_2m}^* \left|i_1j_1\right\rangle \left\langle i_2j_2\right| \\
&= \begin{pmatrix} |\alpha_{000}|^2 + |\alpha_{001}|^2 & \alpha_{000}\alpha_{010}^* + \alpha_{001}\alpha_{011}^* & \alpha_{000}\alpha_{100}^* + \alpha_{001}\alpha_{101}^* & \alpha_{000}\alpha_{110}^* + \alpha_{001}\alpha_{111}^* \\ \alpha_{010}\alpha_{000}^* + \alpha_{011}\alpha_{001}^* & |\alpha_{010}|^2 + |\alpha_{011}|^2 & \alpha_{010}\alpha_{100}^* + \alpha_{011}\alpha_{101}^* & \alpha_{010}\alpha_{110}^* + \alpha_{011}\alpha_{111}^* \\ \alpha_{100}\alpha_{000}^* + \alpha_{101}\alpha_{001}^* & \alpha_{100}\alpha_{010}^* + \alpha_{101}\alpha_{011}^* & |\alpha_{100}|^2 + |\alpha_{101}|^2 & \alpha_{100}\alpha_{110}^* + \alpha_{101}\alpha_{111}^* \\ \alpha_{110}\alpha_{000}^* + \alpha_{111}\alpha_{001}^* & \alpha_{110}\alpha_{010}^* + \alpha_{111}\alpha_{011}^* & \alpha_{110}\alpha_{100}^* + \alpha_{111}\alpha_{101}^* & |\alpha_{110}|^2 + |\alpha_{111}|^2 \end{pmatrix}.
\end{aligned} \tag{A.14}
$$

Matriks densitas tereduksi $\rho_{AC}^{(ent)}$

$$
\begin{aligned}
\rho_{AC}^{(ent)} &= \sum_{m=0}^{1} \sum_{i_1,k_1=0}^{1} \sum_{i_2,k_2=0}^{1} \alpha_{i_1mk_1}\alpha_{i_2mk_2}^* \left|i_1k_1\right\rangle \left\langle i_2k_2\right| \\
&= \begin{pmatrix} |\alpha_{000}|^2 + |\alpha_{010}|^2 & \alpha_{000}\alpha_{001}^* + \alpha_{010}\alpha_{011}^* & \alpha_{000}\alpha_{100}^* + \alpha_{010}\alpha_{110}^* & \alpha_{000}\alpha_{101}^* + \alpha_{010}\alpha_{111}^* \\ \alpha_{001}\alpha_{000}^* + \alpha_{011}\alpha_{010}^* & |\alpha_{001}|^2 + |\alpha_{011}|^2 & \alpha_{001}\alpha_{100}^* + \alpha_{011}\alpha_{110}^* & \alpha_{001}\alpha_{101}^* + \alpha_{011}\alpha_{111}^* \\ \alpha_{100}\alpha_{000}^* + \alpha_{110}\alpha_{010}^* & \alpha_{100}\alpha_{001}^* + \alpha_{110}\alpha_{011}^* & |\alpha_{100}|^2 + |\alpha_{110}|^2 & \alpha_{100}\alpha_{101}^* + \alpha_{110}\alpha_{111}^* \\ \alpha_{101}\alpha_{000}^* + \alpha_{111}\alpha_{010}^* & \alpha_{101}\alpha_{001}^* + \alpha_{111}\alpha_{011}^* & \alpha_{101}\alpha_{100}^* + \alpha_{111}\alpha_{110}^* & |\alpha_{101}|^2 + |\alpha_{111}|^2 \end{pmatrix}.
\end{aligned} \tag{A.15}
$$

Matriks densitas tereduksi $\rho_{BC}^{(ent)}$

$$\begin{aligned}
\rho_{BC}^{(ent)} &= \sum_{m=0}^{1}\sum_{j_1,k_1=0}^{1}\sum_{j_2,k_2=0}^{1} \alpha_{mj_1k_1}\alpha^*_{mj_2k_2} \left|j_1k_1\right\rangle \left\langle j_2k_2\right| \\
&= \begin{pmatrix}
|\alpha_{000}|^2 + |\alpha_{100}|^2 & \alpha_{000}\alpha^*_{001} + \alpha_{100}\alpha^*_{101} & \alpha_{000}\alpha^*_{010} + \alpha_{100}\alpha^*_{110} & \alpha_{000}\alpha^*_{011} + \alpha_{100}\alpha^*_{111} \\
\alpha_{001}\alpha^*_{000} + \alpha_{101}\alpha^*_{100} & |\alpha_{001}|^2 + |\alpha_{101}|^2 & \alpha_{001}\alpha^*_{010} + \alpha_{101}\alpha^*_{110} & \alpha_{001}\alpha^*_{011} + \alpha_{101}\alpha^*_{111} \\
\alpha_{010}\alpha^*_{000} + \alpha_{110}\alpha^*_{100} & \alpha_{010}\alpha^*_{001} + \alpha_{110}\alpha^*_{101} & |\alpha_{010}|^2 + |\alpha_{110}|^2 & \alpha_{010}\alpha^*_{011} + \alpha_{110}\alpha^*_{111} \\
\alpha_{011}\alpha^*_{000} + \alpha_{111}\alpha^*_{100} & \alpha_{011}\alpha^*_{001} + \alpha_{111}\alpha^*_{101} & \alpha_{011}\alpha^*_{010} + \alpha_{111}\alpha^*_{110} & |\alpha_{011}|^2 + |\alpha_{111}|^2
\end{pmatrix}.
\end{aligned} \tag{A.16}$$

A.2.3 Keadaan Gabungan

Keadaan Gabungan A-BC

Untuk keadaan gabungan A-BC, persamaan nilai koefisien nya bisa dituliskan sebagai berikut

$$\alpha_{ijk} = a_i \beta_{jk} \tag{A.17}$$

Matriks densitas tereduksi $\rho_A^{(cp1)}$

$$\begin{aligned}
\rho_A^{(cp1)} &= \sum_{m,n=0}^{1} \sum_{i_1,i_2=0}^{1} a_{i_1}\beta_{mn}a_{i_2}^*\beta_{mn}^* \left|i_1\right\rangle \left\langle i_2\right| \\
&= \begin{pmatrix} a_0\beta_{00}a_0^*\beta_{00}^* + a_0\beta_{01}a_0^*\beta_{01}^* + a_0\beta_{10}a_0^*\beta_{10}^* + a_0\beta_{11}a_0^*\beta_{11}^* & a_0\beta_{00}a_1^*\beta_{00}^* + a_0\beta_{01}a_1^*\beta_{01}^* + a_0\beta_{10}a_1^*\beta_{10}^* + a_0\beta_{11}a_1^*\beta_{11}^* \\ a_1\beta_{00}a_0^*\beta_{00}^* + a_1\beta_{01}a_0^*\beta_{01}^* + a_1\beta_{10}a_0^*\beta_{10}^* + a_1\beta_{11}a_0^*\beta_{11}^* & a_1\beta_{00}a_1^*\beta_{00}^* + a_1\beta_{01}a_1^*\beta_{01}^* + a_1\beta_{10}a_1^*\beta_{10}^* + a_1\beta_{11}a_1^*\beta_{11}^* \end{pmatrix} \\
&= \begin{pmatrix} |a_0|^2\left(|\beta_{00}|^2 + |\beta_{01}|^2 + |\beta_{10}|^2 + |\beta_{11}|^2\right) & a_0a_1^*\left(|\beta_{00}|^2 + |\beta_{01}|^2 + |\beta_{10}|^2 + |\beta_{11}|^2\right) \\ a_1a_0^*\left(|\beta_{00}|^2 + |\beta_{01}|^2 + |\beta_{10}|^2 + |\beta_{11}|^2\right) & |a_1|^2\left(|\beta_{00}|^2 + |\beta_{01}|^2 + |\beta_{10}|^2 + |\beta_{11}|^2\right) \end{pmatrix} \\
\rho_A^{(cp1)} &= \begin{pmatrix} |a_0|^2 & a_0a_1^* \\ a_1a_0^* & |a_1|^2 \end{pmatrix}.
\end{aligned} \tag{A.18}$$

Matriks densitas tereduksi $\rho_B^{(cp1)}$

$$\begin{aligned}
\rho_B^{(cp1)} &= \sum_{m,n=0}^{1} \sum_{j_1,j_2=0}^{1} a_m\beta_{j_1n}a_m^*\beta_{j_2n}^* \left|j_1\right\rangle \left\langle j_2\right| \\
&= \begin{pmatrix} a_0\beta_{00}a_0^*\beta_{00}^* + a_0\beta_{01}a_0^*\beta_{01}^* + a_1\beta_{00}a_1^*\beta_{00}^* + a_1\beta_{01}a_1^*\beta_{01}^* & a_0\beta_{00}a_0^*\beta_{10}^* + a_0\beta_{01}a_0^*\beta_{11}^* + a_1\beta_{00}a_1^*\beta_{10}^* + a_1\beta_{01}a_1^*\beta_{11}^* \\ a_0\beta_{10}a_0^*\beta_{00}^* + a_0\beta_{11}a_0^*\beta_{01}^* + a_1\beta_{10}a_1^*\beta_{00}^* + a_1\beta_{11}a_1^*\beta_{01}^* & a_0\beta_{10}a_0^*\beta_{10}^* + a_0\beta_{11}a_0^*\beta_{11}^* + a_1\beta_{10}a_1^*\beta_{10}^* + a_1\beta_{11}a_1^*\beta_{11}^* \end{pmatrix} \\
&= \begin{pmatrix} \left(|a_0|^2 + |a_1|^2\right)\left(|\beta_{00}|^2 + |\beta_{01}|^2\right) & \left(|a_0|^2 + |a_1|^2\right)\left(\beta_{00}\beta_{10}^* + \beta_{01}\beta_{11}^*\right) \\ \left(|a_0|^2 + |a_1|^2\right)\left(\beta_{10}\beta_{00}^* + \beta_{11}\beta_{01}^*\right) & \left(|a_0|^2 + |a_1|^2\right)\left(|\beta_{10}|^2 + |\beta_{11}|^2\right) \end{pmatrix} \\
\rho_B^{(cp1)} &= \begin{pmatrix} |\beta_{00}|^2 + |\beta_{01}|^2 & \beta_{00}\beta_{10}^* + \beta_{01}\beta_{11}^* \\ \beta_{10}\beta_{00}^* + \beta_{11}\beta_{01}^* & |\beta_{10}|^2 + |\beta_{11}|^2 \end{pmatrix}.
\end{aligned} \tag{A.19}$$

Matriks densitas tereduksi $\rho_C^{(cp1)}$

$$\begin{aligned}
\rho_C^{(cp1)} &= \sum_{m,n=0}^{1}\sum_{k_1,k_2=0}^{1} a_m\beta_{nk_1}a_m^*\beta_{nk_2}^* \left|k_1\right\rangle\left\langle k_2\right| \\
&= \begin{pmatrix} a_0\beta_{00}a_0^*\beta_{00}^* + a_0\beta_{10}a_0^*\beta_{10}^* + a_1\beta_{00}a_1^*\beta_{00}^* + a_1\beta_{10}a_1^*\beta_{10}^* & a_0\beta_{00}a_0^*\beta_{01}^* + a_0\beta_{10}a_0^*\beta_{11}^* + a_1\beta_{00}a_1^*\beta_{01}^* + a_1\beta_{10}a_1^*\beta_{11}^* \\ a_0\beta_{01}a_0^*\beta_{00}^* + a_0\beta_{11}a_0^*\beta_{10}^* + a_1\beta_{01}a_1^*\beta_{00}^* + a_1\beta_{11}a_1^*\beta_{10}^* & a_0\beta_{01}a_0^*\beta_{01}^* + a_0\beta_{11}a_0^*\beta_{11}^* + a_1\beta_{01}a_1^*\beta_{01}^* + a_1\beta_{11}a_1^*\beta_{11}^* \end{pmatrix} \\
&= \begin{pmatrix} \left(|a_0|^2+|a_1|^2\right)\left(|\beta_{00}|^2+|\beta_{10}|^2\right) & \left(|a_0|^2+|a_1|^2\right)\left(\beta_{00}\beta_{01}^*+\beta_{10}\beta_{11}^*\right) \\ \left(|a_0|^2+|a_1|^2\right)\left(\beta_{01}\beta_{00}^*+\beta_{11}\beta_{10}^*\right) & \left(|a_0|^2+|a_1|^2\right)\left(|\beta_{01}|^2+|\beta_{11}|^2\right) \end{pmatrix} \\
\rho_C^{(cp1)} &= \begin{pmatrix} |\beta_{00}|^2+|\beta_{10}|^2 & \beta_{00}\beta_{01}^*+\beta_{10}\beta_{11}^* \\ \beta_{01}\beta_{00}^*+\beta_{11}\beta_{10}^* & |\beta_{01}|^2+|\beta_{11}|^2 \end{pmatrix}.
\end{aligned} \tag{A.20}$$

Matriks densitas tereduksi $\rho_{AB}^{(cp1)}$

$$\begin{aligned}
\rho_{AB}^{(cp1)} &= \sum_{m=0}^{1}\sum_{i_1,j_1=0}^{1}\sum_{i_2,j_2=0}^{1} a_{i_1}\beta_{j_1m}a_{i_2}^*\beta_{j_2m}^* \left|i_1j_1\right\rangle\left\langle i_2j_2\right| \\
&= \begin{pmatrix} a_0\beta_{00}a_0^*\beta_{00}^* + a_0\beta_{01}a_0^*\beta_{01}^* & a_0\beta_{00}a_0^*\beta_{10}^* + a_0\beta_{01}a_0^*\beta_{11}^* & a_0\beta_{00}a_1^*\beta_{00}^* + a_0\beta_{01}a_1^*\beta_{01}^* & a_0\beta_{00}a_1^*\beta_{10}^* + a_0\beta_{01}a_1^*\beta_{11}^* \\ a_0\beta_{10}a_0^*\beta_{00}^* + a_0\beta_{11}a_0^*\beta_{01}^* & a_0\beta_{10}a_0^*\beta_{10}^* + a_0\beta_{11}a_0^*\beta_{11}^* & a_0\beta_{10}a_1^*\beta_{00}^* + a_0\beta_{11}a_1^*\beta_{01}^* & a_0\beta_{10}a_1^*\beta_{10}^* + a_0\beta_{11}a_1^*\beta_{11}^* \\ a_1\beta_{00}a_0^*\beta_{00}^* + a_1\beta_{01}a_0^*\beta_{01}^* & a_1\beta_{00}a_0^*\beta_{10}^* + a_1\beta_{01}a_0^*\beta_{11}^* & a_1\beta_{00}a_1^*\beta_{00}^* + a_1\beta_{01}a_1^*\beta_{01}^* & a_1\beta_{00}a_1^*\beta_{10}^* + a_1\beta_{01}a_1^*\beta_{11}^* \\ a_1\beta_{10}a_0^*\beta_{00}^* + a_1\beta_{11}a_0^*\beta_{01}^* & a_1\beta_{10}a_0^*\beta_{10}^* + a_1\beta_{11}a_0^*\beta_{11}^* & a_1\beta_{10}a_1^*\beta_{00}^* + a_1\beta_{11}a_1^*\beta_{01}^* & a_1\beta_{10}a_1^*\beta_{10}^* + a_1\beta_{11}a_1^*\beta_{11}^* \end{pmatrix} \\
\rho_{AB}^{(cp1)} &= \begin{pmatrix} |a_0|^2\left(|\beta_{00}|^2+|\beta_{01}|^2\right) & |a_0|^2\left(\beta_{00}\beta_{10}^*+\beta_{01}\beta_{11}^*\right) & a_0a_1^*\left(|\beta_{00}|^2+|\beta_{01}|^2\right) & a_0a_1^*\left(\beta_{00}\beta_{10}^*+\beta_{01}\beta_{11}^*\right) \\ |a_0|^2\left(\beta_{10}\beta_{00}^*+\beta_{11}\beta_{01}^*\right) & |a_0|^2\left(|\beta_{10}|^2+|\beta_{11}|^2\right) & a_0a_1^*\left(\beta_{10}\beta_{00}^*+\beta_{11}\beta_{01}^*\right) & a_0a_1^*\left(|\beta_{10}|^2+|\beta_{11}|^2\right) \\ a_1a_0^*\left(|\beta_{00}|^2+|\beta_{01}|^2\right) & a_1a_0^*\left(\beta_{00}\beta_{10}^*+\beta_{01}\beta_{11}^*\right) & |a_1|^2\left(|\beta_{00}|^2+|\beta_{01}|^2\right) & |a_1|^2\left(\beta_{00}\beta_{10}^*+\beta_{01}\beta_{11}^*\right) \\ a_1a_0^*\left(\beta_{10}\beta_{00}^*+\beta_{11}\beta_{01}^*\right) & a_1a_0^*\left(|\beta_{10}|^2+|\beta_{11}|^2\right) & |a_1|^2\left(\beta_{00}\beta_{10}^*+\beta_{01}\beta_{11}^*\right) & |a_1|^2\left(|\beta_{10}|^2+|\beta_{11}|^2\right) \end{pmatrix}.
\end{aligned} \tag{A.21}$$

Matriks densitas tereduksi $\rho_{AC}^{(cp1)}$

$$\rho_{AC}^{(cp1)} = \sum_{m=0}^{1}\sum_{i_1,k_1=0}^{1}\sum_{i_2,k_2=0}^{1} a_{i_1}\beta_{mk_1}a_{i_2}^*\beta_{mk_2}^* \left|i_1k_1\right\rangle\left\langle i_2k_2\right|$$

$$
= \begin{pmatrix}
a_0\beta_{00}a_0^*\beta_{00}^* + a_0\beta_{10}a_0^*\beta_{10}^* & a_0\beta_{00}a_0^*\beta_{01}^* + a_0\beta_{10}a_0^*\beta_{11}^* & a_0\beta_{00}a_1^*\beta_{00}^* + a_0\beta_{10}a_1^*\beta_{10}^* & a_0\beta_{00}a_1^*\beta_{01}^* + a_0\beta_{10}a_1^*\beta_{11}^* \\
a_0\beta_{01}a_0^*\beta_{00}^* + a_0\beta_{11}a_0^*\beta_{10}^* & a_0\beta_{01}a_0^*\beta_{01}^* + a_0\beta_{11}a_0^*\beta_{11}^* & a_0\beta_{01}a_1^*\beta_{00}^* + a_0\beta_{11}a_1^*\beta_{10}^* & a_0\beta_{01}a_1^*\beta_{01}^* + a_0\beta_{11}a_1^*\beta_{11}^* \\
a_1\beta_{00}a_0^*\beta_{00}^* + a_1\beta_{10}a_0^*\beta_{10}^* & a_1\beta_{00}a_0^*\beta_{01}^* + a_1\beta_{10}a_0^*\beta_{11}^* & a_1\beta_{00}a_1^*\beta_{00}^* + a_1\beta_{10}a_1^*\beta_{10}^* & a_1\beta_{00}a_1^*\beta_{01}^* + a_1\beta_{10}a_1^*\beta_{11}^* \\
a_1\beta_{01}a_0^*\beta_{00}^* + a_1\beta_{11}a_0^*\beta_{10}^* & a_1\beta_{01}a_0^*\beta_{01}^* + a_1\beta_{11}a_0^*\beta_{11}^* & a_1\beta_{01}a_1^*\beta_{00}^* + a_1\beta_{11}a_1^*\beta_{10}^* & a_1\beta_{01}a_1^*\beta_{01}^* + a_1\beta_{11}a_1^*\beta_{11}^*
\end{pmatrix}
$$

$$
\rho_{AC}^{(cp1)} = \begin{pmatrix}
|a_0|^2\left(|\beta_{00}|^2 + |\beta_{10}|^2\right) & |a_0|^2\left(\beta_{00}\beta_{01}^* + \beta_{10}\beta_{11}^*\right) & a_0a_1^*\left(|\beta_{00}|^2 + |\beta_{10}|^2\right) & a_0a_1^*\left(\beta_{00}\beta_{01}^* + \beta_{10}\beta_{11}^*\right) \\
|a_0|^2\left(\beta_{01}\beta_{00}^* + \beta_{11}\beta_{10}^*\right) & |a_0|^2\left(|\beta_{01}|^2 + |\beta_{11}|^2\right) & a_0a_1^*\left(\beta_{01}\beta_{00}^* + \beta_{11}\beta_{10}^*\right) & a_0a_1^*\left(|\beta_{01}|^2 + |\beta_{11}|^2\right) \\
a_1a_0^*\left(|\beta_{00}|^2 + |\beta_{10}|^2\right) & a_1a_0^*\left(\beta_{00}\beta_{01}^* + \beta_{10}\beta_{11}^*\right) & |a_1|^2\left(|\beta_{00}|^2 + |\beta_{10}|^2\right) & |a_1|^2\left(\beta_{00}\beta_{01}^* + \beta_{10}\beta_{11}^*\right) \\
a_1a_0^*\left(\beta_{01}\beta_{00}^* + \beta_{11}\beta_{10}^*\right) & a_1a_0^*\left(|\beta_{01}|^2 + |\beta_{11}|^2\right) & |a_1|^2\left(\beta_{01}\beta_{00}^* + \beta_{11}\beta_{10}^*\right) & |a_1|^2\left(|\beta_{01}|^2 + |\beta_{11}|^2\right)
\end{pmatrix}. \tag{A.22}
$$

Matriks densitas tereduksi $\rho_{BC}^{(cp1)}$

$$
\rho_{BC}^{(cp1)} = \sum_{m=0}^{1}\sum_{j_1,k_1=0}^{1}\sum_{j_2,k_2=0}^{1} a_m\beta_{j_1k_1}a_m^*\beta_{j_2k_2}^* \left|j_1k_1\right\rangle \left\langle j_2k_2\right|
$$

$$
= \begin{pmatrix}
a_0\beta_{00}a_0^*\beta_{00}^* + a_1\beta_{00}a_1^*\beta_{00}^* & a_0\beta_{00}a_0^*\beta_{01}^* + a_1\beta_{00}a_1^*\beta_{01}^* & a_0\beta_{00}a_0^*\beta_{10}^* + a_1\beta_{00}a_1^*\beta_{10}^* & a_0\beta_{00}a_0^*\beta_{11}^* + a_1\beta_{00}a_1^*\beta_{11}^* \\
a_0\beta_{01}a_0^*\beta_{00}^* + a_1\beta_{01}a_1^*\beta_{00}^* & a_0\beta_{01}a_0^*\beta_{01}^* + a_1\beta_{01}a_1^*\beta_{01}^* & a_0\beta_{01}a_0^*\beta_{10}^* + a_1\beta_{01}a_1^*\beta_{10}^* & a_0\beta_{01}a_0^*\beta_{11}^* + a_1\beta_{01}a_1^*\beta_{11}^* \\
a_0\beta_{10}a_0^*\beta_{00}^* + a_1\beta_{10}a_1^*\beta_{00}^* & a_0\beta_{10}a_0^*\beta_{01}^* + a_1\beta_{10}a_1^*\beta_{01}^* & a_0\beta_{10}a_0^*\beta_{10}^* + a_1\beta_{10}a_1^*\beta_{10}^* & a_0\beta_{10}a_0^*\beta_{11}^* + a_1\beta_{10}a_1^*\beta_{11}^* \\
a_0\beta_{11}a_0^*\beta_{00}^* + a_1\beta_{11}a_1^*\beta_{00}^* & a_0\beta_{11}a_0^*\beta_{01}^* + a_1\beta_{11}a_1^*\beta_{01}^* & a_0\beta_{11}a_0^*\beta_{10}^* + a_1\beta_{11}a_1^*\beta_{10}^* & a_0\beta_{11}a_0^*\beta_{11}^* + a_1\beta_{11}a_1^*\beta_{11}^*
\end{pmatrix}
$$

$$
= \begin{pmatrix}
\left(|a_0|^2 + |a_1|^2\right)|\beta_{00}|^2 & \left(|a_0|^2 + |a_1|^2\right)\beta_{00}\beta_{01}^* & \left(|a_0|^2 + |a_1|^2\right)\beta_{00}\beta_{10}^* & \left(|a_0|^2 + |a_1|^2\right)\beta_{00}\beta_{11}^* \\
\left(|a_0|^2 + |a_1|^2\right)\beta_{01}\beta_{00}^* & \left(|a_0|^2 + |a_1|^2\right)|\beta_{01}|^2 & \left(|a_0|^2 + |a_1|^2\right)\beta_{01}\beta_{10}^* & \left(|a_0|^2 + |a_1|^2\right)\beta_{01}\beta_{11}^* \\
\left(|a_0|^2 + |a_1|^2\right)\beta_{10}\beta_{00}^* & \left(|a_0|^2 + |a_1|^2\right)\beta_{10}\beta_{01}^* & \left(|a_0|^2 + |a_1|^2\right)|\beta_{10}|^2 & \left(|a_0|^2 + |a_1|^2\right)\beta_{10}\beta_{11}^* \\
\left(|a_0|^2 + |a_1|^2\right)\beta_{11}\beta_{00}^* & \left(|a_0|^2 + |a_1|^2\right)\beta_{11}\beta_{01}^* & \left(|a_0|^2 + |a_1|^2\right)\beta_{11}\beta_{10}^* & \left(|a_0|^2 + |a_1|^2\right)|\beta_{11}|^2
\end{pmatrix}
$$

$$
\rho_{BC}^{(cp1)} = \begin{pmatrix}
|\beta_{00}|^2 & \beta_{00}\beta_{01}^* & \beta_{00}\beta_{10}^* & \beta_{00}\beta_{11}^* \\
\beta_{01}\beta_{00}^* & |\beta_{01}|^2 & \beta_{01}\beta_{10}^* & \beta_{01}\beta_{11}^* \\
\beta_{10}\beta_{00}^* & \beta_{10}\beta_{01}^* & |\beta_{10}|^2 & \beta_{10}\beta_{11}^* \\
\beta_{11}\beta_{00}^* & \beta_{11}\beta_{01}^* & \beta_{11}\beta_{10}^* & |\beta_{11}|^2
\end{pmatrix}. \tag{A.23}
$$

Keadaan Gabungan AB-C

Untuk keadaan gabungan AB-C, persamaan nilai koefisien nya bisa dituliskan sebagai berikut

$$\alpha_{ijk} = \eta_{ij} c_k \tag{A.24}$$

Matriks densitas tereduksi $\rho_A^{(cp2)}$

$$\begin{aligned}
\rho_A^{(cp2)} &= \sum_{m,n=0}^{1} \sum_{i_1,i_2=0}^{1} \eta_{i_1 m} c_n \eta_{i_2 m}^* c_n^* \left| i_1 \right\rangle \left\langle i_2 \right| \\
&= \begin{pmatrix} \eta_{00}c_0\eta_{00}^*c_0^* + \eta_{00}c_1\eta_{00}^*c_1^* + \eta_{01}c_0\eta_{01}^*c_0^* + \eta_{01}c_1\eta_{01}^*c_1^* & \eta_{00}c_0\eta_{10}^*c_0^* + \eta_{00}c_1\eta_{10}^*c_1^* + \eta_{01}c_0\eta_{11}^*c_0^* + \eta_{01}c_1\eta_{11}^*c_1^* \\ \eta_{10}c_0\eta_{00}^*c_0^* + \eta_{10}c_1\eta_{00}^*c_1^* + \eta_{11}c_0\eta_{01}^*c_0^* + \eta_{11}c_1\eta_{01}^*c_1^* & \eta_{10}c_0\eta_{10}^*c_0^* + \eta_{10}c_1\eta_{10}^*c_1^* + \eta_{11}c_0\eta_{11}^*c_0^* + \eta_{11}c_1\eta_{11}^*c_1^* \end{pmatrix} \\
&= \begin{pmatrix} \left(|\eta_{00}|^2 + |\eta_{01}|^2\right)\left(|c_0|^2 + |c_1|^2\right) & \left(\eta_{00}\eta_{10}^* + \eta_{01}\eta_{11}^*\right)\left(|c_0|^2 + |c_1|^2\right) \\ \left(\eta_{10}\eta_{00}^* + \eta_{11}\eta_{01}^*\right)\left(|c_0|^2 + |c_1|^2\right) & \left(|\eta_{10}|^2 + |\eta_{11}|^2\right)\left(|c_0|^2 + |c_1|^2\right) \end{pmatrix} \\
\rho_A^{(cp2)} &= \begin{pmatrix} |\eta_{00}|^2 + |\eta_{01}|^2 & \eta_{00}\eta_{10}^* + \eta_{01}\eta_{11}^* \\ \eta_{10}\eta_{00}^* + \eta_{11}\eta_{01}^* & |\eta_{10}|^2 + |\eta_{11}|^2 \end{pmatrix}.
\end{aligned} \tag{A.25}$$

Matriks densitas tereduksi $\rho_B^{(cp2)}$

$$\begin{aligned}
\rho_B^{(cp2)} &= \sum_{m,n=0}^{1} \sum_{j_1,j_2=0}^{1} \eta_{m j_1} c_n \eta_{m j_2}^* c_n^* \left| j_1 \right\rangle \left\langle j_2 \right| \\
&= \begin{pmatrix} \eta_{00}c_0\eta_{00}^*c_0^* + \eta_{00}c_1\eta_{00}^*c_1^* + \eta_{10}c_0\eta_{10}^*c_0^* + \eta_{10}c_1\eta_{10}^*c_1^* & \eta_{00}c_0\eta_{01}^*c_0^* + \eta_{00}c_1\eta_{01}^*c_1^* + \eta_{10}c_0\eta_{11}^*c_0^* + \eta_{10}c_1\eta_{11}^*c_1^* \\ \eta_{01}c_0\eta_{00}^*c_0^* + \eta_{01}c_1\eta_{00}^*c_1^* + \eta_{11}c_0\eta_{10}^*c_0^* + \eta_{11}c_1\eta_{10}^*c_1^* & \eta_{01}c_0\eta_{01}^*c_0^* + \eta_{01}c_1\eta_{01}^*c_1^* + \eta_{11}c_0\eta_{11}^*c_0^* + \eta_{11}c_1\eta_{11}^*c_1^* \end{pmatrix} \\
&= \begin{pmatrix} \left(|\eta_{00}|^2 + |\eta_{10}|^2\right)\left(|c_0|^2 + |c_1|^2\right) & \left(\eta_{00}\eta_{01}^* + \eta_{10}\eta_{11}^*\right)\left(|c_0|^2 + |c_1|^2\right) \\ \left(\eta_{01}\eta_{00}^* + \eta_{11}\eta_{10}^*\right)\left(|c_0|^2 + |c_1|^2\right) & \left(|\eta_{01}|^2 + |\eta_{11}|^2\right)\left(|c_0|^2 + |c_1|^2\right) \end{pmatrix} \\
\rho_B^{(cp2)} &= \begin{pmatrix} |\eta_{00}|^2 + |\eta_{10}|^2 & \eta_{00}\eta_{01}^* + \eta_{10}\eta_{11}^* \\ \eta_{01}\eta_{00}^* + \eta_{11}\eta_{10}^* & |\eta_{01}|^2 + |\eta_{11}|^2 \end{pmatrix}.
\end{aligned} \tag{A.26}$$

Matriks densitas tereduksi $\rho_C^{(cp2)}$

$$
\begin{aligned}
\rho_C^{(cp2)} &= \sum_{m,n=0}^{1} \sum_{k_1,k_2=0}^{1} \eta_{mn} c_{k_1} \eta_{mn}^* c_{k_2}^* \left|k_1\right\rangle \left\langle k_2\right| \\
&= \begin{pmatrix} \eta_{00}c_0\eta_{00}^*c_0^* + \eta_{01}c_0\eta_{01}^*c_0^* + \eta_{10}c_0\eta_{10}^*c_0^* + \eta_{11}c_0\eta_{11}^*c_0^* & \eta_{00}c_0\eta_{00}^*c_1^* + \eta_{01}c_0\eta_{01}^*c_1^* + \eta_{10}c_0\eta_{10}^*c_1^* + \eta_{11}c_0\eta_{11}^*c_1^* \\ \eta_{00}c_1\eta_{00}^*c_0^* + \eta_{01}c_1\eta_{01}^*c_0^* + \eta_{10}c_1\eta_{10}^*c_0^* + \eta_{11}c_1\eta_{11}^*c_0^* & \eta_{00}c_1\eta_{00}^*c_1^* + \eta_{01}c_1\eta_{01}^*c_1^* + \eta_{10}c_1\eta_{10}^*c_1^* + \eta_{11}c_1\eta_{11}^*c_1^* \end{pmatrix} \\
&= \begin{pmatrix} \left(|\eta_{00}|^2 + |\eta_{01}|^2 + |\eta_{10}|^2 + |\eta_{11}|^2\right)|c_0|^2 & \left(|\eta_{00}|^2 + |\eta_{01}|^2 + |\eta_{10}|^2 + |\eta_{11}|^2\right)c_0c_1^* \\ \left(|\eta_{00}|^2 + |\eta_{01}|^2 + |\eta_{10}|^2 + |\eta_{11}|^2\right)c_1c_0^* & \left(|\eta_{00}|^2 + |\eta_{01}|^2 + |\eta_{10}|^2 + |\eta_{11}|^2\right)|c_1|^2 \end{pmatrix} \\
\rho_C^{(cp2)} &= \begin{pmatrix} |c_0|^2 & c_0c_1^* \\ c_1c_0^* & |c_1|^2 \end{pmatrix}.
\end{aligned}
\tag{A.27}
$$

Matriks densitas tereduksi $\rho_{AB}^{(cp2)}$

$$
\begin{aligned}
\rho_{AB}^{(cp2)} &= \sum_{m=0}^{1} \sum_{i_1,j_1=0}^{1} \sum_{i_2,j_2=0}^{1} \eta_{i_1j_1} c_m \eta_{i_2j_2}^* c_m^* \left|i_1j_1\right\rangle \left\langle i_2j_2\right| \\
&= \begin{pmatrix} \eta_{00}c_0\eta_{00}^*c_0^* + \eta_{00}c_1\eta_{00}^*c_1^* & \eta_{00}c_0\eta_{01}^*c_0^* + \eta_{00}c_1\eta_{01}^*c_1^* & \eta_{00}c_0\eta_{10}^*c_0^* + \eta_{00}c_1\eta_{10}^*c_1^* & \eta_{00}c_0\eta_{11}^*c_0^* + \eta_{00}c_1\eta_{11}^*c_1^* \\ \eta_{01}c_0\eta_{00}^*c_0^* + \eta_{01}c_1\eta_{00}^*c_1^* & \eta_{01}c_0\eta_{01}^*c_0^* + \eta_{01}c_1\eta_{01}^*c_1^* & \eta_{01}c_0\eta_{10}^*c_0^* + \eta_{01}c_1\eta_{10}^*c_1^* & \eta_{01}c_0\eta_{11}^*c_0^* + \eta_{01}c_1\eta_{11}^*c_1^* \\ \eta_{10}c_0\eta_{00}^*c_0^* + \eta_{10}c_1\eta_{00}^*c_1^* & \eta_{10}c_0\eta_{01}^*c_0^* + \eta_{10}c_1\eta_{01}^*c_1^* & \eta_{10}c_0\eta_{10}^*c_0^* + \eta_{10}c_1\eta_{10}^*c_1^* & \eta_{10}c_0\eta_{11}^*c_0^* + \eta_{10}c_1\eta_{11}^*c_1^* \\ \eta_{11}c_0\eta_{00}^*c_0^* + \eta_{11}c_1\eta_{00}^*c_1^* & \eta_{11}c_0\eta_{01}^*c_0^* + \eta_{11}c_1\eta_{01}^*c_1^* & \eta_{11}c_0\eta_{10}^*c_0^* + \eta_{11}c_1\eta_{10}^*c_1^* & \eta_{11}c_0\eta_{11}^*c_0^* + \eta_{11}c_1\eta_{11}^*c_1^* \end{pmatrix} \\
&= \begin{pmatrix} |\eta_{00}|^2\left(|c_0|^2 + |c_1|^2\right) & \eta_{00}\eta_{01}^*\left(|c_0|^2 + |c_1|^2\right) & \eta_{00}\eta_{10}^*\left(|c_0|^2 + |c_1|^2\right) & \eta_{00}\eta_{11}^*\left(|c_0|^2 + |c_1|^2\right) \\ \eta_{01}\eta_{00}^*\left(|c_0|^2 + |c_1|^2\right) & |\eta_{01}|^2\left(|c_0|^2 + |c_1|^2\right) & \eta_{01}\eta_{10}^*\left(|c_0|^2 + |c_1|^2\right) & \eta_{01}\eta_{11}^*\left(|c_0|^2 + |c_1|^2\right) \\ \eta_{10}\eta_{00}^*\left(|c_0|^2 + |c_1|^2\right) & \eta_{10}\eta_{01}^*\left(|c_0|^2 + |c_1|^2\right) & |\eta_{10}|^2\left(|c_0|^2 + |c_1|^2\right) & \eta_{10}\eta_{11}^*\left(|c_0|^2 + |c_1|^2\right) \\ \eta_{11}\eta_{00}^*\left(|c_0|^2 + |c_1|^2\right) & \eta_{11}\eta_{01}^*\left(|c_0|^2 + |c_1|^2\right) & \eta_{11}\eta_{10}^*\left(|c_0|^2 + |c_1|^2\right) & |\eta_{11}|^2\left(|c_0|^2 + |c_1|^2\right) \end{pmatrix}
\end{aligned}
$$

$$\rho_{AB}^{(cp2)} = \begin{pmatrix} |\eta_{00}|^2 & \eta_{00}\eta_{01}^* & \eta_{00}\eta_{10}^* & \eta_{00}\eta_{11}^* \\ \eta_{01}\eta_{00}^* & |\eta_{01}|^2 & \eta_{01}\eta_{10}^* & \eta_{01}\eta_{11}^* \\ \eta_{10}\eta_{00}^* & \eta_{10}\eta_{01}^* & |\eta_{10}|^2 & \eta_{10}\eta_{11}^* \\ \eta_{11}\eta_{00}^* & \eta_{11}\eta_{01}^* & \eta_{11}\eta_{10}^* & |\eta_{11}|^2 \end{pmatrix}. \tag{A.28}$$

Matriks densitas tereduksi $\rho_{AC}^{(cp2)}$

$$\begin{aligned}\rho_{AC}^{(cp2)} &= \sum_{m=0}^{1}\sum_{i_1,k_1=0}^{1}\sum_{i_2,k_2=0}^{1} \eta_{i_1 m} c_{k_1} \eta_{i_2 m}^* c_{k_2}^* \left|i_1 k_1\right\rangle \left\langle i_2 k_2\right| \\ &= \begin{pmatrix} \eta_{00}c_0\eta_{00}^*c_0^* + \eta_{01}c_0\eta_{01}^*c_0^* & \eta_{00}c_0\eta_{00}^*c_1^* + \eta_{01}c_0\eta_{01}^*c_1^* & \eta_{00}c_1\eta_{00}^*c_0^* + \eta_{01}c_1\eta_{01}^*c_0^* & \eta_{00}c_1\eta_{01}^*c_1^* + \eta_{01}c_1\eta_{01}^*c_1^* \\ \eta_{00}c_0\eta_{10}^*c_0^* + \eta_{01}c_0\eta_{11}^*c_0^* & \eta_{00}c_0\eta_{10}^*c_1^* + \eta_{01}c_0\eta_{11}^*c_1^* & \eta_{00}c_1\eta_{10}^*c_0^* + \eta_{01}c_1\eta_{11}^*c_0^* & \eta_{00}c_1\eta_{11}^*c_1^* + \eta_{01}c_1\eta_{11}^*c_1^* \\ \eta_{10}c_0\eta_{00}^*c_0^* + \eta_{11}c_0\eta_{01}^*c_0^* & \eta_{10}c_0\eta_{00}^*c_1^* + \eta_{11}c_0\eta_{01}^*c_1^* & \eta_{10}c_1\eta_{00}^*c_0^* + \eta_{11}c_1\eta_{01}^*c_0^* & \eta_{10}c_1\eta_{01}^*c_1^* + \eta_{11}c_1\eta_{01}^*c_1^* \\ \eta_{10}c_0\eta_{10}^*c_0^* + \eta_{11}c_0\eta_{11}^*c_0^* & \eta_{10}c_0\eta_{10}^*c_1^* + \eta_{11}c_0\eta_{11}^*c_1^* & \eta_{10}c_1\eta_{10}^*c_0^* + \eta_{11}c_1\eta_{11}^*c_0^* & \eta_{10}c_1\eta_{11}^*c_1^* + \eta_{11}c_1\eta_{11}^*c_1^* \end{pmatrix}\end{aligned}$$

$$\rho_{AC}^{(cp2)} = \begin{pmatrix} \left(|\eta_{00}|^2 + |\eta_{01}|^2\right)|c_0|^2 & \left(|\eta_{00}|^2 + |\eta_{01}|^2\right)c_0c_1^* & \left(|\eta_{00}|^2 + |\eta_{01}|^2\right)c_1c_0^* & \left(|\eta_{00}|^2 + |\eta_{01}|^2\right)|c_1|^2 \\ \left(\eta_{00}\eta_{10}^* + \eta_{01}\eta_{11}^*\right)|c_0|^2 & \left(\eta_{00}\eta_{10}^* + \eta_{01}\eta_{11}^*\right)c_0c_1^* & \left(\eta_{00}\eta_{10}^* + \eta_{01}\eta_{11}^*\right)c_1c_0^* & \left(\eta_{00}\eta_{10}^* + \eta_{01}\eta_{11}^*\right)|c_1|^2 \\ \left(\eta_{10}\eta_{00}^* + \eta_{11}\eta_{01}^*\right)|c_0|^2 & \left(\eta_{10}\eta_{00}^* + \eta_{11}\eta_{01}^*\right)c_0c_1^* & \left(\eta_{10}\eta_{00}^* + \eta_{11}\eta_{01}^*\right)c_1c_0^* & \left(\eta_{10}\eta_{00}^* + \eta_{11}\eta_{01}^*\right)|c_1|^2 \\ \left(|\eta_{10}|^2 + |\eta_{11}|^2\right)|c_0|^2 & \left(|\eta_{10}|^2 + |\eta_{11}|^2\right)c_0c_1^* & \left(|\eta_{10}|^2 + |\eta_{11}|^2\right)c_1c_0^* & \left(|\eta_{10}|^2 + |\eta_{11}|^2\right)|c_1|^2 \end{pmatrix}. \tag{A.29}$$

Matriks densitas tereduksi $\rho_{BC}^{(cp2)}$

$$\begin{aligned}\rho_{BC}^{(cp2)} &= \sum_{m=0}^{1}\sum_{j_1,k_1=0}^{1}\sum_{j_2,k_2=0}^{1} \eta_{mj_1} c_{k_1} \eta_{mj_2}^* c_{k_2}^* \left|j_1 k_1\right\rangle \left\langle j_2 k_2\right| \\ &= \begin{pmatrix} \eta_{00}c_0\eta_{00}^*c_0^* + \eta_{10}c_0\eta_{10}^*c_0^* & \eta_{00}c_0\eta_{00}^*c_1^* + \eta_{10}c_0\eta_{10}^*c_1^* & \eta_{00}c_1\eta_{00}^*c_0^* + \eta_{10}c_1\eta_{10}^*c_0^* & \eta_{00}c_1\eta_{00}^*c_1^* + \eta_{10}c_1\eta_{10}^*c_1^* \\ \eta_{00}c_0\eta_{01}^*c_0^* + \eta_{10}c_0\eta_{11}^*c_0^* & \eta_{00}c_0\eta_{01}^*c_1^* + \eta_{10}c_0\eta_{11}^*c_1^* & \eta_{00}c_1\eta_{01}^*c_0^* + \eta_{10}c_1\eta_{11}^*c_0^* & \eta_{00}c_1\eta_{01}^*c_1^* + \eta_{10}c_1\eta_{11}^*c_1^* \\ \eta_{01}c_0\eta_{00}^*c_0^* + \eta_{11}c_0\eta_{10}^*c_0^* & \eta_{01}c_0\eta_{00}^*c_1^* + \eta_{11}c_0\eta_{10}^*c_1^* & \eta_{01}c_1\eta_{00}^*c_0^* + \eta_{11}c_1\eta_{10}^*c_0^* & \eta_{01}c_1\eta_{00}^*c_1^* + \eta_{11}c_1\eta_{10}^*c_1^* \\ \eta_{01}c_0\eta_{01}^*c_0^* + \eta_{11}c_0\eta_{11}^*c_0^* & \eta_{01}c_0\eta_{01}^*c_1^* + \eta_{11}c_0\eta_{11}^*c_1^* & \eta_{01}c_1\eta_{01}^*c_0^* + \eta_{11}c_1\eta_{11}^*c_0^* & \eta_{01}c_1\eta_{01}^*c_1^* + \eta_{11}c_1\eta_{11}^*c_1^* \end{pmatrix}\end{aligned}$$

$$\rho_{BC}^{(cp2)} = \begin{pmatrix} \left(|\eta_{00}|^2 + |\eta_{10}|^2\right) |c_0|^2 & \left(|\eta_{00}|^2 + |\eta_{10}|^2\right) c_0 c_1^* & \left(|\eta_{00}|^2 + |\eta_{10}|^2\right) c_1 c_0^* & \left(|\eta_{00}|^2 + |\eta_{10}|^2\right) |c_1|^2 \\ (\eta_{00}\eta_{01}^* + \eta_{10}\eta_{11}^*) |c_0|^2 & (\eta_{00}\eta_{01}^* + \eta_{10}\eta_{11}^*) c_0 c_1^* & (\eta_{00}\eta_{01}^* + \eta_{10}\eta_{11}^*) c_1 c_0^* & (\eta_{00}\eta_{01}^* + \eta_{10}\eta_{11}^*) |c_1|^2 \\ (\eta_{01}\eta_{00}^* + \eta_{11}\eta_{10}^*) |c_0|^2 & (\eta_{01}\eta_{00}^* + \eta_{11}\eta_{10}^*) c_0 c_1^* & (\eta_{01}\eta_{00}^* + \eta_{11}\eta_{10}^*) c_1 c_0^* & (\eta_{01}\eta_{00}^* + \eta_{11}\eta_{10}^*) |c_1|^2 \\ \left(|\eta_{01}|^2 + |\eta_{11}|^2\right) |c_0|^2 & \left(|\eta_{01}|^2 + |\eta_{11}|^2\right) c_0 c_1^* & \left(|\eta_{01}|^2 + |\eta_{11}|^2\right) c_1 c_0^* & \left(|\eta_{01}|^2 + |\eta_{11}|^2\right) |c_1|^2 \end{pmatrix}. \tag{A.30}$$

Keadaan Gabungan AC-B

Untuk keadaan gabungan AC-B, persamaan nilai koefisien nya bisa dituliskan sebagai berikut

$$\alpha_{ijk} = b_j \zeta_{ik} \tag{A.31}$$

Matriks densitas tereduksi $\rho_A^{(cp3)}$

$$\begin{aligned}
\rho_A^{(cp3)} &= \sum_{m,n=0}^{1} \sum_{i_1,i_2=0}^{1} b_m \zeta_{i_1 n} b_m^* \zeta_{i_2 n}^* \left| i_1 \right\rangle \left\langle i_2 \right| \\
&= \begin{pmatrix} b_0\zeta_{00}b_0^*\zeta_{00}^* + b_0\zeta_{01}b_0^*\zeta_{01}^* + b_1\zeta_{00}b_1^*\zeta_{00}^* + b_1\zeta_{01}b_1^*\zeta_{01}^* & b_0\zeta_{00}b_0^*\zeta_{10}^* + b_0\zeta_{01}b_0^*\zeta_{11}^* + b_1\zeta_{00}b_1^*\zeta_{10}^* + b_1\zeta_{01}b_1^*\zeta_{11}^* \\ b_0\zeta_{10}b_0^*\zeta_{00}^* + b_0\zeta_{11}b_0^*\zeta_{01}^* + b_1\zeta_{10}b_1^*\zeta_{00}^* + b_1\zeta_{11}b_1^*\zeta_{01}^* & b_0\zeta_{10}b_0^*\zeta_{10}^* + b_0\zeta_{11}b_0^*\zeta_{11}^* + b_1\zeta_{10}b_1^*\zeta_{10}^* + b_1\zeta_{11}b_1^*\zeta_{11}^* \end{pmatrix} \\
&= \begin{pmatrix} \left(|b_0|^2 + |b_1|^2\right)\left(|\zeta_{00}|^2 + |\zeta_{01}|^2\right) & \left(|b_0|^2 + |b_1|^2\right)\left(\zeta_{00}\zeta_{10}^* + \zeta_{01}\zeta_{11}^*\right) \\ \left(|b_0|^2 + |b_1|^2\right)\left(\zeta_{10}\zeta_{00}^* + \zeta_{11}\zeta_{01}^*\right) & \left(|b_0|^2 + |b_1|^2\right)\left(|\zeta_{10}|^2 + |\zeta_{11}|^2\right) \end{pmatrix} \\
\rho_A^{(cp3)} &= \begin{pmatrix} |\zeta_{00}|^2 + |\zeta_{01}|^2 & \zeta_{00}\zeta_{10}^* + \zeta_{01}\zeta_{11}^* \\ \zeta_{10}\zeta_{00}^* + \zeta_{11}\zeta_{01}^* & |\zeta_{10}|^2 + |\zeta_{11}|^2 \end{pmatrix}.
\end{aligned} \tag{A.32}$$

Matriks densitas tereduksi $\rho_B^{(cp3)}$

$$\begin{aligned}
\rho_B^{(cp3)} &= \sum_{m,n=0}^{1} \sum_{j_1,j_2=0}^{1} b_{j_1} \zeta_{mn} b_{j_2}^* \zeta_{mn}^* \left| j_1 \right\rangle \left\langle j_2 \right| \\
&= \begin{pmatrix} b_0\zeta_{00}b_0^*\zeta_{00}^* + b_0\zeta_{01}b_0^*\zeta_{01}^* + b_0\zeta_{10}b_0^*\zeta_{10}^* + b_0\zeta_{11}b_0^*\zeta_{11}^* & b_0\zeta_{00}b_1^*\zeta_{00}^* + b_0\zeta_{01}b_1^*\zeta_{01}^* + b_0\zeta_{10}b_1^*\zeta_{10}^* + b_0\zeta_{11}b_1^*\zeta_{11}^* \\ b_1\zeta_{00}b_0^*\zeta_{00}^* + b_1\zeta_{01}b_0^*\zeta_{01}^* + b_1\zeta_{10}b_0^*\zeta_{10}^* + b_1\zeta_{11}b_0^*\zeta_{11}^* & b_1\zeta_{00}b_1^*\zeta_{00}^* + b_1\zeta_{01}b_1^*\zeta_{01}^* + b_1\zeta_{10}b_1^*\zeta_{10}^* + b_1\zeta_{11}b_1^*\zeta_{11}^* \end{pmatrix} \\
&= \begin{pmatrix} |b_0|^2\left(|\zeta_{00}|^2 + |\zeta_{01}|^2 + |\zeta_{10}|^2 + |\zeta_{11}|^2\right) & b_0b_1^*\left(|\zeta_{00}|^2 + |\zeta_{01}|^2 + |\zeta_{10}|^2 + |\zeta_{11}|^2\right) \\ b_1b_0^*\left(|\zeta_{00}|^2 + |\zeta_{01}|^2 + |\zeta_{10}|^2 + |\zeta_{11}|^2\right) & |b_1|^2\left(|\zeta_{00}|^2 + |\zeta_{01}|^2 + |\zeta_{10}|^2 + |\zeta_{11}|^2\right) \end{pmatrix} \\
\rho_B^{(cp3)} &= \begin{pmatrix} |b_0|^2 & b_0b_1^* \\ b_1b_0^* & |b_1|^2 \end{pmatrix}.
\end{aligned} \tag{A.33}$$

Matriks densitas tereduksi $\rho_C^{(cp3)}$

$$\rho_C^{(cp3)} = \sum_{m,n=0}^{1}\sum_{k_1,k_2=0}^{1} b_n\zeta_{mk_1}b_n^*\zeta_{mk_2}^* \left|k_1\right\rangle\left\langle k_2\right|$$

$$= \begin{pmatrix} b_0\zeta_{00}b_0^*\zeta_{00}^* + b_1\zeta_{00}b_1^*\zeta_{00}^* + b_0\zeta_{10}b_0^*\zeta_{10}^* + b_1\zeta_{10}b_1^*\zeta_{10}^* & b_0\zeta_{00}b_0^*\zeta_{01}^* + b_1\zeta_{00}b_1^*\zeta_{01}^* + b_0\zeta_{10}b_0^*\zeta_{11}^* + b_1\zeta_{10}b_1^*\zeta_{11}^* \\ b_0\zeta_{01}b_0^*\zeta_{00}^* + b_1\zeta_{01}b_1^*\zeta_{00}^* + b_0\zeta_{11}b_0^*\zeta_{10}^* + b_1\zeta_{11}b_1^*\zeta_{10}^* & b_0\zeta_{01}b_0^*\zeta_{01}^* + b_1\zeta_{01}b_1^*\zeta_{01}^* + b_0\zeta_{11}b_0^*\zeta_{11}^* + b_1\zeta_{11}b_1^*\zeta_{11}^* \end{pmatrix}$$

$$= \begin{pmatrix} \left(\left|b_0\right|^2 + \left|b_1\right|^2\right)\left(\left|\zeta_{00}\right|^2 + \left|\zeta_{10}\right|^2\right) & \left(\left|b_0\right|^2 + \left|b_1\right|^2\right)\left(\zeta_{00}\zeta_{01}^* + \zeta_{10}\zeta_{11}^*\right) \\ \left(\left|b_0\right|^2 + \left|b_1\right|^2\right)\left(\zeta_{01}\zeta_{00}^* + \zeta_{11}\zeta_{10}^*\right) & \left(\left|b_0\right|^2 + \left|b_1\right|^2\right)\left(\left|\zeta_{01}\right|^2 + \left|\zeta_{11}\right|^2\right) \end{pmatrix}$$

$$\rho_C^{(cp3)} = \begin{pmatrix} \left|\zeta_{00}\right|^2 + \left|\zeta_{10}\right|^2 & \zeta_{00}\zeta_{01}^* + \zeta_{10}\zeta_{11}^* \\ \zeta_{01}\zeta_{00}^* + \zeta_{11}\zeta_{10}^* & \left|\zeta_{01}\right|^2 + \left|\zeta_{11}\right|^2 \end{pmatrix}. \tag{A.34}$$

Matriks densitas tereduksi $\rho_{AB}^{(cp3)}$

$$\rho_{AB}^{(cp3)} = \sum_{m=0}^{1}\sum_{i_1,j_1=0}^{1}\sum_{i_2,j_2=0}^{1} b_{j_1}\zeta_{i_1m}b_{j_2}^*\zeta_{i_2m}^* \left|i_1j_1\right\rangle\left\langle i_2j_2\right|$$

$$= \begin{pmatrix} b_0\zeta_{00}b_0^*\zeta_{00}^* + b_0\zeta_{01}b_0^*\zeta_{01}^* & b_0\zeta_{00}b_1^*\zeta_{00}^* + b_0\zeta_{01}b_1^*\zeta_{01}^* & b_0\zeta_{00}b_0^*\zeta_{10}^* + b_0\zeta_{01}b_0^*\zeta_{11}^* & b_0\zeta_{00}b_1^*\zeta_{10}^* + b_0\zeta_{01}b_1^*\zeta_{11}^* \\ b_1\zeta_{00}b_0^*\zeta_{00}^* + b_1\zeta_{01}b_0^*\zeta_{01}^* & b_1\zeta_{00}b_1^*\zeta_{00}^* + b_1\zeta_{01}b_1^*\zeta_{01}^* & b_1\zeta_{00}b_0^*\zeta_{10}^* + b_1\zeta_{01}b_0^*\zeta_{11}^* & b_1\zeta_{00}b_1^*\zeta_{10}^* + b_1\zeta_{01}b_1^*\zeta_{11}^* \\ b_0\zeta_{10}b_0^*\zeta_{00}^* + b_0\zeta_{11}b_0^*\zeta_{01}^* & b_0\zeta_{10}b_1^*\zeta_{00}^* + b_0\zeta_{11}b_1^*\zeta_{01}^* & b_0\zeta_{10}b_0^*\zeta_{10}^* + b_0\zeta_{11}b_0^*\zeta_{11}^* & b_0\zeta_{10}b_1^*\zeta_{10}^* + b_0\zeta_{11}b_1^*\zeta_{11}^* \\ b_1\zeta_{10}b_0^*\zeta_{00}^* + b_1\zeta_{11}b_0^*\zeta_{01}^* & b_1\zeta_{10}b_1^*\zeta_{00}^* + b_1\zeta_{11}b_1^*\zeta_{01}^* & b_1\zeta_{10}b_0^*\zeta_{10}^* + b_1\zeta_{11}b_0^*\zeta_{11}^* & b_1\zeta_{10}b_1^*\zeta_{10}^* + b_1\zeta_{11}b_1^*\zeta_{11}^* \end{pmatrix}$$

$$\rho_{AB}^{(cp3)} = \begin{pmatrix} \left|b_0\right|^2\left(\left|\zeta_{00}\right|^2 + \left|\zeta_{01}\right|^2\right) & b_0b_1^*\left(\left|\zeta_{00}\right|^2 + \left|\zeta_{01}\right|^2\right) & \left|b_0\right|^2\left(\zeta_{00}\zeta_{10}^* + \zeta_{01}\zeta_{11}^*\right) & b_0b_1^*\left(\zeta_{00}\zeta_{10}^* + \zeta_{01}\zeta_{11}^*\right) \\ b_1b_0^*\left(\left|\zeta_{00}\right|^2 + \left|\zeta_{01}\right|^2\right) & \left|b_1\right|^2\left(\left|\zeta_{00}\right|^2 + \left|\zeta_{01}\right|^2\right) & b_1b_0^*\left(\zeta_{00}\zeta_{10}^* + \zeta_{01}\zeta_{11}^*\right) & \left|b_1\right|^2\left(\zeta_{00}\zeta_{10}^* + \zeta_{01}\zeta_{11}^*\right) \\ \left|b_0\right|^2\left(\zeta_{10}\zeta_{00}^* + \zeta_{11}\zeta_{01}^*\right) & b_0b_1^*\left(\zeta_{10}\zeta_{00}^* + \zeta_{11}\zeta_{01}^*\right) & \left|b_0\right|^2\left(\left|\zeta_{10}\right|^2 + \left|\zeta_{11}\right|^2\right) & b_0b_1^*\left(\left|\zeta_{10}\right|^2 + \left|\zeta_{11}\right|^2\right) \\ b_1b_0^*\left(\zeta_{10}\zeta_{00}^* + \zeta_{11}\zeta_{01}^*\right) & \left|b_1\right|^2\left(\zeta_{10}\zeta_{00}^* + \zeta_{11}\zeta_{01}^*\right) & b_1b_0^*\left(\left|\zeta_{10}\right|^2 + \left|\zeta_{11}\right|^2\right) & \left|b_1\right|^2\left(\left|\zeta_{10}\right|^2 + \left|\zeta_{11}\right|^2\right) \end{pmatrix}. \tag{A.35}$$

Matriks densitas tereduksi $\rho_{AC}^{(cp3)}$

$$\rho_{AC}^{(cp3)} = \sum_{m=0}^{1}\sum_{i_1,k_1=0}^{1}\sum_{i_2,k_2=0}^{1} b_m\zeta_{i_1k_1}b_m^*\zeta_{i_2k_2}^* \left|i_1k_1\right\rangle\left\langle i_2k_2\right|$$

$$
= \begin{pmatrix} b_0\zeta_{00}b_0^*\zeta_{00}^* + b_1\zeta_{00}b_1^*\zeta_{00}^* & b_0\zeta_{00}b_0^*\zeta_{01}^* + b_1\zeta_{00}b_1^*\zeta_{01}^* & b_0\zeta_{00}b_0^*\zeta_{10}^* + b_1\zeta_{00}b_1^*\zeta_{10}^* & b_0\zeta_{00}b_0^*\zeta_{11}^* + b_1\zeta_{00}b_1^*\zeta_{11}^* \\ b_0\zeta_{01}b_0^*\zeta_{00}^* + b_1\zeta_{01}b_1^*\zeta_{00}^* & b_0\zeta_{01}b_0^*\zeta_{01}^* + b_1\zeta_{01}b_1^*\zeta_{01}^* & b_0\zeta_{01}b_0^*\zeta_{10}^* + b_1\zeta_{01}b_1^*\zeta_{10}^* & b_0\zeta_{01}b_0^*\zeta_{11}^* + b_1\zeta_{01}b_1^*\zeta_{11}^* \\ b_0\zeta_{10}b_0^*\zeta_{00}^* + b_1\zeta_{10}b_1^*\zeta_{00}^* & b_0\zeta_{10}b_0^*\zeta_{01}^* + b_1\zeta_{10}b_1^*\zeta_{01}^* & b_0\zeta_{10}b_0^*\zeta_{10}^* + b_1\zeta_{10}b_1^*\zeta_{10}^* & b_0\zeta_{10}b_0^*\zeta_{11}^* + b_1\zeta_{10}b_1^*\zeta_{11}^* \\ b_0\zeta_{11}b_0^*\zeta_{00}^* + b_1\zeta_{11}b_1^*\zeta_{00}^* & b_0\zeta_{11}b_0^*\zeta_{01}^* + b_1\zeta_{11}b_1^*\zeta_{01}^* & b_0\zeta_{11}b_0^*\zeta_{10}^* + b_1\zeta_{11}b_1^*\zeta_{10}^* & b_0\zeta_{11}b_0^*\zeta_{11}^* + b_1\zeta_{11}b_1^*\zeta_{11}^* \end{pmatrix}
$$

$$
= \begin{pmatrix} \left(|b_0|^2 + |b_1|^2\right)|\zeta_{00}|^2 & \left(|b_0|^2 + |b_1|^2\right)\zeta_{00}\zeta_{01}^* & \left(|b_0|^2 + |b_1|^2\right)\zeta_{00}\zeta_{10}^* & \left(|b_0|^2 + |b_1|^2\right)\zeta_{00}\zeta_{11}^* \\ \left(|b_0|^2 + |b_1|^2\right)\zeta_{01}\zeta_{00}^* & \left(|b_0|^2 + |b_1|^2\right)|\zeta_{01}|^2 & \left(|b_0|^2 + |b_1|^2\right)\zeta_{01}\zeta_{10}^* & \left(|b_0|^2 + |b_1|^2\right)\zeta_{01}\zeta_{11}^* \\ \left(|b_0|^2 + |b_1|^2\right)\zeta_{10}\zeta_{00}^* & \left(|b_0|^2 + |b_1|^2\right)\zeta_{10}\zeta_{01}^* & \left(|b_0|^2 + |b_1|^2\right)|\zeta_{10}|^2 & \left(|b_0|^2 + |b_1|^2\right)\zeta_{10}\zeta_{11}^* \\ \left(|b_0|^2 + |b_1|^2\right)\zeta_{11}\zeta_{00}^* & \left(|b_0|^2 + |b_1|^2\right)\zeta_{11}\zeta_{01}^* & \left(|b_0|^2 + |b_1|^2\right)\zeta_{11}\zeta_{10}^* & \left(|b_0|^2 + |b_1|^2\right)|\zeta_{11}|^2 \end{pmatrix}
$$

$$
\rho_{AC}^{(cp3)} = \begin{pmatrix} |\zeta_{00}|^2 & \zeta_{00}\zeta_{01}^* & \zeta_{00}\zeta_{10}^* & \zeta_{00}\zeta_{11}^* \\ \zeta_{01}\zeta_{00}^* & |\zeta_{01}|^2 & \zeta_{01}\zeta_{10}^* & \zeta_{01}\zeta_{11}^* \\ \zeta_{10}\zeta_{00}^* & \zeta_{10}\zeta_{01}^* & |\zeta_{10}|^2 & \zeta_{10}\zeta_{11}^* \\ \zeta_{11}\zeta_{00}^* & \zeta_{11}\zeta_{01}^* & \zeta_{11}\zeta_{10}^* & |\zeta_{11}|^2 \end{pmatrix}. \tag{A.36}
$$

Matriks densitas tereduksi $\rho_{BC}^{(cp3)}$

$$
\rho_{BC}^{(cp3)} = \sum_{m=0}^{1}\sum_{j_1,k_1=0}^{1}\sum_{j_2,k_2=0}^{1} b_{j_1}\zeta_{mk_1}b_{j_2}^*\zeta_{mk_2}^* \left|j_1k_1\right\rangle \left\langle j_2k_2\right|
$$

$$
= \begin{pmatrix} b_0\zeta_{00}b_0^*\zeta_{00}^* + b_0\zeta_{10}b_0^*\zeta_{10}^* & b_0\zeta_{00}b_0^*\zeta_{01}^* + b_0\zeta_{10}b_0^*\zeta_{11}^* & b_0\zeta_{00}b_1^*\zeta_{00}^* + b_0\zeta_{10}b_1^*\zeta_{10}^* & b_0\zeta_{00}b_1^*\zeta_{01}^* + b_0\zeta_{10}b_1^*\zeta_{11}^* \\ b_0\zeta_{01}b_0^*\zeta_{00}^* + b_0\zeta_{11}b_0^*\zeta_{10}^* & b_0\zeta_{01}b_0^*\zeta_{01}^* + b_0\zeta_{11}b_0^*\zeta_{11}^* & b_0\zeta_{01}b_1^*\zeta_{00}^* + b_0\zeta_{11}b_1^*\zeta_{10}^* & b_0\zeta_{01}b_1^*\zeta_{01}^* + b_0\zeta_{11}b_1^*\zeta_{11}^* \\ b_1\zeta_{00}b_0^*\zeta_{00}^* + b_1\zeta_{10}b_0^*\zeta_{10}^* & b_1\zeta_{00}b_0^*\zeta_{01}^* + b_1\zeta_{10}b_0^*\zeta_{11}^* & b_1\zeta_{00}b_1^*\zeta_{00}^* + b_1\zeta_{10}b_1^*\zeta_{10}^* & b_1\zeta_{00}b_1^*\zeta_{01}^* + b_1\zeta_{10}b_1^*\zeta_{11}^* \\ b_1\zeta_{01}b_0^*\zeta_{00}^* + b_1\zeta_{11}b_0^*\zeta_{10}^* & b_1\zeta_{01}b_0^*\zeta_{01}^* + b_1\zeta_{11}b_0^*\zeta_{11}^* & b_1\zeta_{01}b_1^*\zeta_{00}^* + b_1\zeta_{11}b_1^*\zeta_{10}^* & b_1\zeta_{01}b_1^*\zeta_{01}^* + b_1\zeta_{11}b_1^*\zeta_{11}^* \end{pmatrix}
$$

$$
\rho_{BC}^{(cp3)} = \begin{pmatrix} |b_0|^2\left(|\zeta_{00}|^2 + |\zeta_{10}|^2\right) & |b_0|^2\left(\zeta_{00}\zeta_{01}^* + \zeta_{10}\zeta_{11}^*\right) & b_0b_1^*\left(|\zeta_{00}|^2 + |\zeta_{10}|^2\right) & b_0b_1^*\left(\zeta_{00}\zeta_{01}^* + \zeta_{10}\zeta_{11}^*\right) \\ |b_0|^2\left(\zeta_{01}\zeta_{00}^* + \zeta_{11}\zeta_{10}^*\right) & |b_0|^2\left(|\zeta_{01}|^2 + |\zeta_{11}|^2\right) & b_0b_1^*\left(\zeta_{01}\zeta_{00}^* + \zeta_{11}\zeta_{10}^*\right) & b_0b_1^*\left(|\zeta_{01}|^2 + |\zeta_{11}|^2\right) \\ b_1b_0^*\left(|\zeta_{00}|^2 + |\zeta_{10}|^2\right) & b_1b_0^*\left(\zeta_{00}\zeta_{01}^* + \zeta_{10}\zeta_{11}^*\right) & |b_1|^2\left(|\zeta_{00}|^2 + |\zeta_{10}|^2\right) & |b_1|^2\left(\zeta_{00}\zeta_{01}^* + \zeta_{10}\zeta_{11}^*\right) \\ b_1b_0^*\left(\zeta_{01}\zeta_{00}^* + \zeta_{11}\zeta_{10}^*\right) & b_1b_0^*\left(|\zeta_{01}|^2 + |\zeta_{11}|^2\right) & |b_1|^2\left(\zeta_{01}\zeta_{00}^* + \zeta_{11}\zeta_{10}^*\right) & |b_1|^2\left(|\zeta_{01}|^2 + |\zeta_{11}|^2\right) \end{pmatrix}. \tag{A.37}
$$

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Lampiran B

Penurunan Persamaan Entropi Shannon dan Entropi Von Neumann

B.1 Entropi Shannon

Tinjau entropi pada fisika statistik di persamaan (2.104)

$$S = k\left(\ln Z - \beta\frac{\partial \ln Z}{\partial \beta}\right),$$

dengan fungsi partisi $Z = \sum_r e^{-\beta E_r}$ dan reprosikal temperatur $\beta = \frac{1}{k_B T}$. Selanjutnya, bisa dilakukan perhitungan berikut

$$\begin{aligned}
S &= k\left(\ln Z - \beta\frac{\partial Z}{Z\partial \beta}\right)\\
&= k\left(\ln Z - \frac{\beta}{Z}\frac{\partial}{\partial \beta}\left(\sum_r e^{-\beta E_r}\right)\right)\\
&= \frac{k}{Z}\left(Z\ln Z - \beta\sum_r\left(\frac{\partial}{\partial \beta}(e^{-\beta E_r})\right)\right)\\
&= \frac{k}{Z}\left(Z\ln Z + \sum_r \beta E_r e^{-\beta E_r}\right)\\
&= \frac{k}{Z}\left(\sum_r e^{-\beta E_r}\ln Z + \sum_r e^{-\beta E_r}\beta E_r\right)\\
&= -\frac{k}{Z}\sum_r e^{-\beta E_r}\left(-\ln Z - \beta E_r\right)\\
&= -k\sum_r \frac{e^{-\beta E_r}}{Z}\left(\ln e^{-\beta E_r} - \ln Z\right)\\
&= -k\sum_r \left(\frac{e^{-\beta E_r}}{Z}\right)\ln\left(\frac{e^{-\beta E_r}}{Z}\right).
\end{aligned}$$

Suku $\frac{e^{-\beta E_r}}{Z}$ merupakan fungsi distribusi probabilitas dari ensembel kanonik (Purwanto, 2007)

$$p_r = \frac{e^{-\beta E_r}}{Z}, \tag{B.1}$$

sehingga

$$S = -k \sum_r p_r \ln p_r,$$

atau

$$S = -k \sum_i p_i \ln p_i.$$

Jika diambil *natural unit* berupa $k = 1$, maka

$$\therefore S = -\sum_i p_i \ln p_i.$$

Basis logaritma persamaan diatas bisa diganti menjadi 2 agar bisa sama seperti persamaan (2.107).

B.2 Entropi Von Neumann

Tinjau persamaan entropi Von Neumann pada persamaan (2.108)

$$S(\rho) = \text{tr}(\rho \log \rho).$$

Kita bisa tinjau nilai $\log(\rho)$ terlebih dahulu. Mula-mula, tuliskan fungsi logaritma $\log(1+x)$ dalam ekspansi deret Maclaurin

$$\log(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$$

$$\log(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (x-1)^n. \tag{B.2}$$

Maka,

$$\log \rho = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (\rho - 1)^n. \tag{B.3}$$

Deret binomial didefinisikan sebagai

$$(a+b)^n = \sum_{k=0}^{\infty} \binom{n}{k} a^{n-k} b^k, \tag{B.4}$$

sehingga

$$(\rho - 1)^n = \sum_{k=0}^{\infty} \binom{n}{k} \rho^{n-k} (-1)^k. \tag{B.5}$$

Eksponen dari suatu matriks A bisa dicari menggunakan diagonalisasi matriks (2.45)

$$A = PDP^{-1} = P \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_m \end{pmatrix} P^{-1}$$

$$A^n = (PDP^{-1})(PDP^{-1}) \ldots (PDP^{-1})(PDP^{-1}),$$

dan karena $P^{-1}P = PP^{-1} = \mathbb{I}$ sesuai definisi invers pada persamaan (2.25), maka

$$A^n = PD^nP^{-1} = P \begin{pmatrix} \lambda_1^n & & & \\ & \lambda_2^n & & \\ & & \ddots & \\ & & & \lambda_m^n \end{pmatrix} P^{-1}.$$

Maka, ρ^{n-k} bisa dituliskan sebagai

$$\rho^{n-k} = PD^{n-k}P^{-1}.$$

Persamaan (B.5) menjadi

$$\begin{aligned}
(\rho - 1)^n &= \sum_{k=0}^{\infty} \binom{n}{k} \left(PD^{n-k}P^{-1}\right)(-1)^k \\
&= P \left(\sum_{k=0}^{\infty} \binom{n}{k} D^{n-k}(-1)^k \right) P^{-1} \\
&= P \left(\sum_{k=0}^{\infty} \binom{n}{k} \begin{bmatrix} \lambda_1^{n-k} & & & \\ & \lambda_2^{n-k} & & \\ & & \ddots & \\ & & & \lambda_m^{n-k} \end{bmatrix} (-1)^k \right) P^{-1} \\
&= P \left(\begin{bmatrix} \sum_{k=0}^{\infty} \binom{n}{k} \lambda_1^{n-k}(-1)^k & & & \\ & \sum_{k=0}^{\infty} \binom{n}{k} \lambda_2^{n-k}(-1)^k & & \\ & & \ddots & \\ & & & \sum_{k=0}^{\infty} \binom{n}{k} \lambda_m^{n-k}(-1)^k \end{bmatrix} \right) P^{-1},
\end{aligned}$$

namun setiap elemen diagonal dari persamaan diatas merupakan persamaan (B.4), dengan $a = \lambda_m^{n-k}$ dan $b = (-1)^k$, sehingga persamaan diatas bisa disederhanakan menjadi

$$(\rho - 1)^n = P \begin{pmatrix} (\lambda_1 - 1)^n & & & \\ & (\lambda_2 - 1)^n & & \\ & & \ddots & \\ & & & (\lambda_m - 1)^n \end{pmatrix} P^{-1}. \tag{B.6}$$

Substitusi persamaan ini kembali ke persamaan (B.3), sehingga menghasilkan

$$\begin{aligned}
\log\rho &= \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}P\begin{pmatrix}(\lambda_1-1)^n & & & \\ & (\lambda_2-1)^n & & \\ & & \ddots & \\ & & & (\lambda_m-1)^n\end{pmatrix}P^{-1}\\
&= P\begin{pmatrix}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}(\lambda_1-1)^n & & & \\ & \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}(\lambda_2-1)^n & & \\ & & \ddots & \\ & & & \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}(\lambda_m-1)^n\end{pmatrix}P^{-1}.
\end{aligned}$$

Masing-masing elemen diagonal sesuai dengan persamaan (B.2) dengan nilai $x=\lambda_m$. Maka persamaan diatas menjadi

$$\log\rho = P\begin{pmatrix}\log\lambda_1 & & & \\ & \log\lambda_2 & & \\ & & \ddots & \\ & & & \log\lambda_m\end{pmatrix}P^{-1}. \tag{B.7}$$

Misalkan matriks Λ sebagai

$$\Lambda = \begin{pmatrix}\log\lambda_1 & & & \\ & \log\lambda_2 & & \\ & & \ddots & \\ & & & \log\lambda_m\end{pmatrix},$$

Sehingga persamaan (2.108) menjadi

$$\begin{aligned}
S(\rho) &= \operatorname{tr}(\rho\log\rho)\\
&= \operatorname{tr}\left(PD\underbrace{P^{-1}P}_{\mathbb{I}}\Lambda P^{-1}\right)\\
&= \operatorname{tr}\big(PD\Lambda P^{-1}\big).
\end{aligned}$$

Trace matriks memenuhi sifat permutasi siklik yang ditunjukkan pada persamaan (2.31), sehingga

$$\begin{aligned}
S(\rho) &= \operatorname{tr}\left(D\Lambda\underbrace{P^{-1}P}_{\mathbb{I}}\right)\\
&= \operatorname{tr}(D\Lambda)\\
&= \operatorname{tr}\left(\begin{pmatrix}\lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_m\end{pmatrix}\begin{pmatrix}\log\lambda_1 & & & \\ & \log\lambda_2 & & \\ & & \ddots & \\ & & & \log\lambda_m\end{pmatrix}\right)
\end{aligned}$$

$$= \operatorname{tr}\left(\begin{pmatrix} \lambda_1 \log \lambda_1 & & & \\ & \lambda_2 \log \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_m \log \lambda_m \end{pmatrix}\right)$$

$$\therefore S(\rho) = \sum_{i=1}^{m} \lambda_i \log \lambda_i.$$

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Lampiran C

Perhitungan Entropi Von Neumann Dua Qubit Dan Tiga Qubit

C.1 Entropi Keadaan Dua Qubit

C.1.1 Keadaan Asal Sistem AB

Persamaan karakteristik matriks densitas asal ρ_{AB} pada persamaan (2.88) adalah

$$|\rho_{AB} - \lambda I| = 0,$$

maka

$$\begin{vmatrix} |\alpha_{00}|^2 - \lambda & \alpha_{00}\alpha_{01}^* & \alpha_{00}\alpha_{10}^* & \alpha_{00}\alpha_{11}^* \\ \alpha_{01}\alpha_{00}^* & |\alpha_{01}|^2 - \lambda & \alpha_{01}\alpha_{10}^* & \alpha_{01}\alpha_{11}^* \\ \alpha_{10}\alpha_{00}^* & \alpha_{10}\alpha_{01}^* & |\alpha_{10}|^2 - \lambda & \alpha_{10}\alpha_{11}^* \\ \alpha_{11}\alpha_{00}^* & \alpha_{11}\alpha_{01}^* & \alpha_{11}\alpha_{10}^* & |\alpha_{11}|^2 - \lambda \end{vmatrix} = 0$$

Kita bisa lakukan eliminasi Gauss untuk menyederhanakan determinan diatas serta melakukan ekspansi kofaktor untuk bisa menghitung nilai determinannya

1. **(OKE)**
   $\boldsymbol{K_2 \to K_2 - \frac{\alpha_{00}\alpha_{01}^*}{|\alpha_{00}|^2-\lambda} K_1}$
   $\boldsymbol{K_3 \to K_3 - \frac{\alpha_{00}\alpha_{10}^*}{|\alpha_{00}|^2-\lambda} K_1}$
   $\boldsymbol{K_4 \to K_4 - \frac{\alpha_{00}\alpha_{11}^*}{|\alpha_{00}|^2-\lambda} K_1}$

$$\begin{vmatrix} |\alpha_{00}|^2 - \lambda & 0 & 0 & 0 \\ \alpha_{01}\alpha_{00}^* & \frac{\lambda(|\alpha_{00}|^2+|\alpha_{01}|^2-\lambda)}{-|\alpha_{00}|^2+\lambda} & \frac{\alpha_{01}\alpha_{10}^*\lambda}{-|\alpha_{00}|^2+\lambda} & \frac{\alpha_{01}\alpha_{11}^*\lambda}{-|\alpha_{00}|^2+\lambda} \\ \alpha_{10}\alpha_{00}^* & \frac{\alpha_{10}\alpha_{01}^*\lambda}{-|\alpha_{00}|^2+\lambda} & \frac{\lambda(|\alpha_{00}|^2+|\alpha_{10}|^2-\lambda)}{-|\alpha_{00}|^2+\lambda} & \frac{\alpha_{10}\alpha_{11}^*\lambda}{-|\alpha_{00}|^2+\lambda} \\ \alpha_{11}\alpha_{00}^* & \frac{\alpha_{11}\alpha_{01}^*\lambda}{-|\alpha_{00}|^2+\lambda} & \frac{\alpha_{11}\alpha_{01}^*\lambda}{-|\alpha_{00}|^2+\lambda} & \frac{\lambda(|\alpha_{00}|^2+|\alpha_{11}|^2-\lambda)}{-|\alpha_{00}|^2+\lambda} \end{vmatrix} = 0$$

2. **(Ekspansi baris ke-1)**

$$(|\alpha_{00}|^2 - \lambda)M_{11} - 0M_{12} + 0M_{13} + 0M_{14} = 0$$

$$(|\alpha_{00}|^2 - \lambda)\begin{vmatrix} \frac{\lambda(|\alpha_{00}|^2+|\alpha_{01}|^2-\lambda)}{-|\alpha_{00}|^2+\lambda} & \frac{\alpha_{01}\alpha_{10}^*\lambda}{-|\alpha_{00}|^2+\lambda} & \frac{\alpha_{01}\alpha_{11}^*\lambda}{-|\alpha_{00}|^2+\lambda} \\ \frac{\alpha_{10}\alpha_{01}^*\lambda}{-|\alpha_{00}|^2+\lambda} & \frac{\lambda(|\alpha_{00}|^2+|\alpha_{10}|^2-\lambda)}{-|\alpha_{00}|^2+\lambda} & \frac{\alpha_{10}\alpha_{11}^*\lambda}{-|\alpha_{00}|^2+\lambda} \\ \frac{\alpha_{11}\alpha_{01}^*\lambda}{-|\alpha_{00}|^2+\lambda} & \frac{\alpha_{11}\alpha_{01}^*\lambda}{-|\alpha_{00}|^2+\lambda} & \frac{\lambda(|\alpha_{00}|^2+|\alpha_{11}|^2-\lambda)}{-|\alpha_{00}|^2+\lambda} \end{vmatrix} = 0$$

3. **(OKE)**
   $\boldsymbol{K_2 \to K_2 - \frac{\alpha_{01}\alpha_{10}^*}{|\alpha_{00}|^2+|\alpha_{01}|^2-\lambda} K_1}$
   $\boldsymbol{K_3 \to K_3 - \frac{\alpha_{01}\alpha_{11}^*}{|\alpha_{00}|^2+|\alpha_{01}|^2-\lambda} K_1}$

$$(|\alpha_{00}|^2 - \lambda)\begin{vmatrix} \frac{\lambda(|\alpha_{00}|^2+|\alpha_{01}|^2-\lambda)}{-|\alpha_{00}|^2+\lambda} & 0 & 0 \\ \frac{\alpha_{10}\alpha_{01}^*\lambda}{-|\alpha_{00}|^2+\lambda} & \frac{\lambda(|\alpha_{00}|^2+|\alpha_{01}|^2+|\alpha_{10}|^2-\lambda)}{-|\alpha_{00}|^2-|\alpha_{01}|^2+\lambda} & \frac{\alpha_{10}\alpha_{11}^*\lambda}{-|\alpha_{00}|^2-|\alpha_{01}|^2+\lambda} \\ \frac{\alpha_{11}\alpha_{01}^*\lambda}{-|\alpha_{00}|^2+\lambda} & \frac{\alpha_{11}\alpha_{01}^*\lambda}{-|\alpha_{00}|^2-|\alpha_{01}|^2+\lambda} & \frac{\lambda(|\alpha_{00}|^2+|\alpha_{01}|^2|\alpha_{11}|^2-\lambda)}{-|\alpha_{00}|^2-|\alpha_{01}|^2+\lambda} \end{vmatrix} = 0$$

4. **(Ekspansi baris ke-1)**

$$(|\alpha_{00}|^2 - \lambda)\left(\frac{\lambda(|\alpha_{00}|^2 + |\alpha_{01}|^2 - \lambda)}{-|\alpha_{00}|^2 + \lambda} M_{11} - 0M_{12} + 0M_{13}\right) = 0$$

$$(|\alpha_{00}|^2 - \lambda)\left(\frac{\lambda(|\alpha_{00}|^2 + |\alpha_{01}|^2 - \lambda)}{-|\alpha_{00}|^2 + \lambda}\right)\begin{vmatrix} \frac{\lambda(|\alpha_{00}|^2+|\alpha_{01}|^2+|\alpha_{10}|^2-\lambda)}{-|\alpha_{00}|^2-|\alpha_{01}|^2+\lambda} & \frac{\alpha_{10}\alpha_{11}^*\lambda}{-|\alpha_{00}|^2-|\alpha_{01}|^2+\lambda} \\ \frac{\alpha_{11}\alpha_{01}^*\lambda}{-|\alpha_{00}|^2-|\alpha_{01}|^2+\lambda} & \frac{\lambda(|\alpha_{00}|^2+|\alpha_{01}|^2|\alpha_{11}|^2-\lambda)}{-|\alpha_{00}|^2-|\alpha_{01}|^2+\lambda} \end{vmatrix} = 0$$

5. **(Determinan 2×2)**

$$(|\alpha_{00}|^2 - \lambda)\left(\frac{\lambda(|\alpha_{00}|^2 + |\alpha_{01}|^2 - \lambda)}{-|\alpha_{00}|^2 + \lambda}\right)\left(\frac{-|\alpha_{00}|^2\lambda^2 - |\alpha_{01}|^2\lambda^2 - |\alpha_{10}|^2\lambda^2 - |\alpha_{11}|^2\lambda^2 + \lambda^3}{-|\alpha_{00}|^2 - |\alpha_{01}|^2 + \lambda}\right) = 0$$

$$\lambda^3(\lambda - (|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2)) = 0$$

Karena koefisien α_{ij} memenuhi syarat normalisasi $\sum_{i,j=0}^{1} |\alpha_{ij}|^2 = 1$, maka persamaan diatas menjadi

$$\begin{gathered} \lambda^3(\lambda - 1) = 0 \\ \lambda_1 = 1 \qquad \lambda_{2,3,4} = 0. \end{gathered} \tag{C.1}$$

C.1.2 Keadaan Terpisah

Persamaan karakteristik matriks densitas tereduksi $\rho_A^{(sep)}$ pada persamaan (A.1) adalah

$$\left|\rho_A^{(sep)} - \lambda I\right| = 0,$$

maka

$$\begin{aligned}
\begin{vmatrix} |a_0|^2 - \lambda & a_0 a_1^* \\ a_1 a_0^* & |a_1|^2 - \lambda \end{vmatrix} &= 0 \\
\left(|a_0|^2 - \lambda\right)\left(|a_1|^2 - \lambda\right) - \left(a_0 a_1^*\right)\left(a_1 a_0^*\right) &= 0 \\
|a_0|^2|a_1|^2 - \left(|a_0|^2 + |a_1|^2\right)\lambda + \lambda^2 - |a_0|^2|a_1|^2 &= 0 \\
\lambda^2 - \lambda &= 0 \\
\lambda_1 = 1 \qquad \lambda_2 &= 0.
\end{aligned} \tag{C.2}$$

Persamaan karakteristik matriks densitas tereduksi $\rho_B^{(sep)}$ pada persamaan (A.2) adalah

$$\left|\rho_B^{(sep)} - \lambda I\right| = 0,$$

maka

$$\begin{aligned}
\begin{vmatrix} |b_0|^2 - \lambda & b_0 b_1^* \\ b_1 b_0^* & |b_1|^2 - \lambda \end{vmatrix} &= 0 \\
\left(|b_0|^2 - \lambda\right)\left(|b_1|^2 - \lambda\right) - \left(b_0 b_1^*\right)\left(b_1 b_0^*\right) &= 0 \\
|b_0|^2|b_1|^2 - \left(|b_0|^2 + |b_1|^2\right)\lambda + \lambda^2 - |b_0|^2|b_1|^2 &= 0 \\
\lambda^2 - \lambda &= 0 \\
\lambda_1 = 1 \qquad \lambda_2 &= 0.
\end{aligned} \tag{C.3}$$

C.1.3 Keadaan Terbelit

Persamaan karakteristik matriks densitas tereduksi $\rho_A^{(ent)}$ pada persamaan (A.3) adalah

$$\left|\rho_A^{(ent)} - \lambda I\right| = 0,$$

maka

$$\begin{aligned}
\begin{vmatrix} |\alpha_{00}|^2 + |\alpha_{01}|^2 - \lambda & \alpha_{00}\alpha_{10}^* + \alpha_{01}\alpha_{11}^* \\ \alpha_{10}\alpha_{00}^* + \alpha_{11}\alpha_{01}^* & |\alpha_{10}|^2 + |\alpha_{11}|^2 - \lambda \end{vmatrix} &= 0 \\
\left(|\alpha_{00}|^2 + |\alpha_{01}|^2 - \lambda\right)\left(|\alpha_{10}|^2 + |\alpha_{11}|^2 - \lambda\right) & \\
- \left(\alpha_{00}\alpha_{10}^* + \alpha_{01}\alpha_{11}^*\right)\left(\alpha_{10}\alpha_{00}^* + \alpha_{11}\alpha_{01}^*\right) &= 0 \\
\left(|\alpha_{00}|^2 + |\alpha_{01}|^2\right)\left(|\alpha_{10}|^2 + |\alpha_{11}|^2\right) & \\
- \left(|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2\right)\lambda + \lambda^2 & \\
- \left(|\alpha_{00}|^2|\alpha_{10}|^2 + \alpha_{00}\alpha_{11}\alpha_{01}^*\alpha_{10}^* + \alpha_{01}\alpha_{10}\alpha_{00}^*\alpha_{11}^* + |\alpha_{01}|^2|\alpha_{11}|^2\right) &= 0 \\
\lambda^2 - \lambda + \left(|\alpha_{00}|^2|\alpha_{11}|^2 + |\alpha_{01}|^2|\alpha_{10}|^2 - \alpha_{00}\alpha_{11}\alpha_{01}^*\alpha_{10}^* - \alpha_{01}\alpha_{10}\alpha_{00}^*\alpha_{11}^*\right) &= 0
\end{aligned}$$

$$\lambda_{1,2} = \frac{1}{2}\left(1 \pm \sqrt{1 - 4\left(|\alpha_{00}|^2|\alpha_{11}|^2 + |\alpha_{01}|^2|\alpha_{10}|^2 - \alpha_{00}\alpha_{11}\alpha_{01}^*\alpha_{10}^* - \alpha_{01}\alpha_{10}\alpha_{00}^*\alpha_{11}^*\right)}\right). \tag{C.4}$$

Persamaan karakteristik matriks densitas tereduksi $\rho_B^{(ent)}$ pada persamaan (A.4) adalah

$$\left|\rho_B^{(ent)} - \lambda I\right| = 0,$$

maka

$$\begin{aligned}
\begin{vmatrix} |\alpha_{00}|^2 + |\alpha_{10}|^2 - \lambda & \alpha_{00}\alpha_{01}^* + \alpha_{10}\alpha_{11}^* \\ \alpha_{01}\alpha_{00}^* + \alpha_{11}\alpha_{10}^* & |\alpha_{01}|^2 + |\alpha_{11}|^2 - \lambda \end{vmatrix} &= 0 \\
\left(|\alpha_{00}|^2 + |\alpha_{10}|^2 - \lambda\right)\left(|\alpha_{01}|^2 + |\alpha_{11}|^2 - \lambda\right) & \\
- \left(\alpha_{00}\alpha_{01}^* + \alpha_{10}\alpha_{11}^*\right)\left(\alpha_{01}\alpha_{00}^* + \alpha_{11}\alpha_{10}^*\right) &= 0 \\
\left(|\alpha_{00}|^2 + |\alpha_{10}|^2\right)\left(|\alpha_{01}|^2 + |\alpha_{11}|^2\right) & \\
- \left(|\alpha_{00}|^2 + |\alpha_{10}|^2 + |\alpha_{01}|^2 + |\alpha_{11}|^2\right)\lambda + \lambda^2 & \\
- \left(|\alpha_{00}|^2|\alpha_{01}|^2 + \alpha_{00}\alpha_{11}\alpha_{01}^*\alpha_{10}^* + \alpha_{01}\alpha_{10}\alpha_{00}^*\alpha_{11}^* + |\alpha_{10}|^2|\alpha_{11}|^2\right) &= 0 \\
\lambda^2 - \lambda + \left(|\alpha_{00}|^2|\alpha_{11}|^2 + |\alpha_{01}|^2|\alpha_{10}|^2 - \alpha_{00}\alpha_{11}\alpha_{01}^*\alpha_{10}^* - \alpha_{01}\alpha_{10}\alpha_{00}^*\alpha_{11}^*\right) &= 0
\end{aligned}$$

$$\lambda_{1,2} = \frac{1}{2}\left(1 \pm \sqrt{1 - 4\left(|\alpha_{00}|^2|\alpha_{11}|^2 + |\alpha_{01}|^2|\alpha_{10}|^2 - \alpha_{00}\alpha_{11}\alpha_{01}^*\alpha_{10}^* - \alpha_{01}\alpha_{10}\alpha_{00}^*\alpha_{11}^*\right)}\right). \tag{C.5}$$

C.2 Entropi Keadaan Tiga Qubit

C.2.1 Keadaan Asal Sistem ABC

Persamaan karakteristik dari persamaan (4.33)

$$|\rho_{ABC} - \lambda I| = 0,$$

maka

$$\begin{vmatrix}
|\alpha_{000}|^2-\lambda & \alpha_{000}\alpha_{001}^* & \alpha_{000}\alpha_{010}^* & \alpha_{000}\alpha_{011}^* & \alpha_{000}\alpha_{100}^* & \alpha_{000}\alpha_{101}^* & \alpha_{000}\alpha_{110}^* & \alpha_{000}\alpha_{111}^* \\
\alpha_{001}\alpha_{000}^* & |\alpha_{001}|^2-\lambda & \alpha_{001}\alpha_{010}^* & \alpha_{001}\alpha_{011}^* & \alpha_{001}\alpha_{100}^* & \alpha_{001}\alpha_{101}^* & \alpha_{001}\alpha_{110}^* & \alpha_{001}\alpha_{111}^* \\
\alpha_{010}\alpha_{000}^* & \alpha_{010}\alpha_{001}^* & |\alpha_{010}|^2-\lambda & \alpha_{010}\alpha_{011}^* & \alpha_{010}\alpha_{100}^* & \alpha_{010}\alpha_{101}^* & \alpha_{010}\alpha_{110}^* & \alpha_{010}\alpha_{111}^* \\
\alpha_{011}\alpha_{000}^* & \alpha_{011}\alpha_{001}^* & \alpha_{011}\alpha_{010}^* & |\alpha_{011}|^2-\lambda & \alpha_{011}\alpha_{100}^* & \alpha_{011}\alpha_{101}^* & \alpha_{011}\alpha_{110}^* & \alpha_{011}\alpha_{111}^* \\
\alpha_{100}\alpha_{000}^* & \alpha_{100}\alpha_{001}^* & \alpha_{100}\alpha_{010}^* & \alpha_{100}\alpha_{011}^* & |\alpha_{100}|^2-\lambda & \alpha_{100}\alpha_{101}^* & \alpha_{100}\alpha_{110}^* & \alpha_{100}\alpha_{111}^* \\
\alpha_{101}\alpha_{000}^* & \alpha_{101}\alpha_{001}^* & \alpha_{101}\alpha_{010}^* & \alpha_{101}\alpha_{011}^* & \alpha_{101}\alpha_{100}^* & |\alpha_{101}|^2-\lambda & \alpha_{101}\alpha_{110}^* & \alpha_{101}\alpha_{111}^* \\
\alpha_{110}\alpha_{000}^* & \alpha_{110}\alpha_{001}^* & \alpha_{110}\alpha_{010}^* & \alpha_{110}\alpha_{011}^* & \alpha_{110}\alpha_{100}^* & \alpha_{110}\alpha_{101}^* & |\alpha_{110}|^2-\lambda & \alpha_{110}\alpha_{111}^* \\
\alpha_{111}\alpha_{000}^* & \alpha_{111}\alpha_{001}^* & \alpha_{111}\alpha_{010}^* & \alpha_{111}\alpha_{011}^* & \alpha_{111}\alpha_{100}^* & \alpha_{111}\alpha_{101}^* & \alpha_{111}\alpha_{110}^* & |\alpha_{111}|^2-\lambda
\end{vmatrix} = 0$$

Lakukan eliminasi Gauss untuk menyederhanakan determinan diatas serta melakukan ekspansi kofaktor untuk bisa menghitung nilai determinannya

1. **(OKE)**
   $\boldsymbol{K_2 \to K_2 - \frac{\alpha_{000}\alpha_{001}^*}{|\alpha_{000}|^2-\lambda} K_1}$
   $\boldsymbol{K_3 \to K_3 - \frac{\alpha_{000}\alpha_{010}^*}{|\alpha_{000}|^2-\lambda} K_1}$
   $\boldsymbol{K_4 \to K_4 - \frac{\alpha_{000}\alpha_{011}^*}{|\alpha_{000}|^2-\lambda} K_1}$
   $\boldsymbol{K_5 \to K_5 - \frac{\alpha_{000}\alpha_{100}^*}{|\alpha_{000}|^2-\lambda} K_1}$
   $\boldsymbol{K_6 \to K_6 - \frac{\alpha_{000}\alpha_{101}^*}{|\alpha_{000}|^2-\lambda} K_1}$
   $\boldsymbol{K_7 \to K_7 - \frac{\alpha_{000}\alpha_{110}^*}{|\alpha_{000}|^2-\lambda} K_1}$

$\boldsymbol{K_8 \rightarrow K_8 - \frac{\alpha_{000}\alpha^*_{111}}{|\alpha_{000}|^2-\lambda} K_1}$

$$\begin{vmatrix}
|\alpha_{000}|^2-\lambda & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\alpha_{001}\alpha^*_{000} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\alpha_{010}\alpha^*_{000} & \frac{\alpha_{010}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{010}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\alpha_{011}\alpha^*_{000} & \frac{\alpha_{011}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{011}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{011}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{011}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{011}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{011}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\alpha_{100}\alpha^*_{000} & \frac{\alpha_{100}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\alpha_{101}\alpha^*_{000} & \frac{\alpha_{101}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\alpha_{110}\alpha^*_{000} & \frac{\alpha_{110}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\alpha_{111}\alpha^*_{000} & \frac{\alpha_{111}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda}
\end{vmatrix} = 0$$

2. **(Ekspansi baris ke-1)**

$$(\mathcal{S}_1)M_{11} - 0M_{12} + 0M_{13} - 0M_{14} + 0M_{15} - 0M_{16} + 0M_{17} - 0M_{18} = 0$$

$$\mathcal{S}_1 \begin{vmatrix}
\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{001}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\frac{\alpha_{010}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{010}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\frac{\alpha_{011}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{011}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{011}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{011}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{011}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{011}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{010}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\frac{\alpha_{100}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\frac{\alpha_{101}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\frac{\alpha_{110}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha^*_{111}\lambda}{-|\alpha_{000}|^2+\lambda} \\
\frac{\alpha_{111}\alpha^*_{001}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha^*_{010}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha^*_{011}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha^*_{100}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha^*_{101}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha^*_{110}\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda}
\end{vmatrix} = 0,$$

dengan $\mathcal{S}_1 = |\alpha_{000}|^2 - \lambda$.

3. **(OKE)**

$$\boldsymbol{K_2 \to K_2 - \frac{\alpha_{001}\alpha_{010}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda} K_1}$$
$$\boldsymbol{K_3 \to K_3 - \frac{\alpha_{001}\alpha_{011}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda} K_1}$$
$$\boldsymbol{K_4 \to K_4 - \frac{\alpha_{001}\alpha_{100}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda} K_1}$$
$$\boldsymbol{K_5 \to K_5 - \frac{\alpha_{001}\alpha_{101}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda} K_1}$$
$$\boldsymbol{K_6 \to K_6 - \frac{\alpha_{001}\alpha_{110}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda} K_1}$$
$$\boldsymbol{K_7 \to K_7 - \frac{\alpha_{001}\alpha_{111}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda} K_1}$$

$$\mathcal{S}_1 \begin{vmatrix}
\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda} & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\alpha_{010}\alpha_{001}^*\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} \\
\frac{\alpha_{011}\alpha_{001}^*\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{011}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{011}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{011}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{011}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{011}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} \\
\frac{\alpha_{100}\alpha_{001}^*\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{100}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{100}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{100}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{100}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{100}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} \\
\frac{\alpha_{101}\alpha_{001}^*\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{101}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{101}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{101}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{101}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{101}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} \\
\frac{\alpha_{110}\alpha_{001}^*\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{110}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{110}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{110}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{110}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} \\
\frac{\alpha_{111}\alpha_{001}^*\lambda}{-|\alpha_{000}|^2+\lambda} & \frac{\alpha_{111}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{111}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{111}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{111}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda}
\end{vmatrix} = 0$$

4. **(Ekspansi baris ke-1)**

$$\mathcal{S}_1\left(\mathcal{S}_2 M_{11} - 0M_{12} + 0M_{13} - 0M_{14} + 0M_{15} - 0M_{16} + 0M_{17}\right) = 0$$

$$\mathcal{S}_1\mathcal{S}_2 \begin{vmatrix}
\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} \\
\frac{\alpha_{011}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{011}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{011}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{011}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{011}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{010}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} \\
\frac{\alpha_{100}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{100}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{100}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{100}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{100}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} \\
\frac{\alpha_{101}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{101}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{101}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{101}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{101}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} \\
\frac{\alpha_{110}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{110}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{110}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{110}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} \\
\frac{\alpha_{111}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{111}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{111}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{111}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda}
\end{vmatrix} = 0,$$

dengan $\mathcal{S}_2 = \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda)}{-|\alpha_{000}|^2+\lambda}$.

5. **(OKE)**

$$\boldsymbol{K_2 \rightarrow K_2 - \frac{\alpha_{010}\alpha_{011}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2-\lambda}K_1}$$
$$\boldsymbol{K_3 \rightarrow K_3 - \frac{\alpha_{010}\alpha_{100}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2-\lambda}K_1}$$
$$\boldsymbol{K_4 \rightarrow K_4 - \frac{\alpha_{010}\alpha_{101}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2-\lambda}K_1}$$
$$\boldsymbol{K_5 \rightarrow K_5 - \frac{\alpha_{010}\alpha_{110}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2-\lambda}K_1}$$
$$\boldsymbol{K_6 \rightarrow K_6 - \frac{\alpha_{010}\alpha_{111}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2-\lambda}K_1}$$

$$\mathcal{S}_1\mathcal{S}_2 \begin{vmatrix}
\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & 0 & 0 & 0 & 0 & 0 \\
\frac{\alpha_{011}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{011}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{011}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{011}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{010}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} \\
\frac{\alpha_{100}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{100}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{100}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{100}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{100}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} \\
\frac{\alpha_{101}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{101}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{101}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{101}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{101}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} \\
\frac{\alpha_{110}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{110}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{110}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{110}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} \\
\frac{\alpha_{111}\alpha_{010}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda} & \frac{\alpha_{111}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{111}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{111}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda}
\end{vmatrix} = 0$$

6. **(Ekspansi baris ke-1)**

$$\mathcal{S}_1\mathcal{S}_2\left(\mathcal{S}_3 M_{11} - 0M_{12} + 0M_{13} - 0M_{14} + 0M_{15} - 0M_{16}\right) = 0$$

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3 \begin{vmatrix}
\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{011}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{011}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{011}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{010}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} \\
\frac{\alpha_{100}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{100}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{100}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{100}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} \\
\frac{\alpha_{101}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{101}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{101}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{101}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} \\
\frac{\alpha_{110}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{110}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{110}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} \\
\frac{\alpha_{111}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{111}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{111}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda}
\end{vmatrix} = 0,$$

dengan $\mathcal{S}_3 = \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda}$.

7. **(OKE)**

$$\boldsymbol{K_2 \to K_2 - \frac{\alpha_{011}\alpha_{100}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2-\lambda} K_1}$$
$$\boldsymbol{K_3 \to K_3 - \frac{\alpha_{011}\alpha_{101}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2-\lambda} K_1}$$
$$\boldsymbol{K_4 \to K_4 - \frac{\alpha_{011}\alpha_{110}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2-\lambda} K_1}$$
$$\boldsymbol{K_5 \to K_5 - \frac{\alpha_{011}\alpha_{111}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2-\lambda} K_1}$$

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3 \begin{vmatrix}
\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & 0 & 0 & 0 & 0 \\
\frac{\alpha_{100}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{100}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{100}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{100}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} \\
\frac{\alpha_{101}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{101}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{101}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{101}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} \\
\frac{\alpha_{110}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{110}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{110}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} \\
\frac{\alpha_{111}\alpha_{011}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda} & \frac{\alpha_{111}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{111}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda}
\end{vmatrix} = 0$$

8. **(Ekspansi baris ke-1)**

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3\left(\mathcal{S}_4 M_{11} - 0M_{12} + 0M_{13} - 0M_{14} + 0M_{15}\right) = 0$$

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3\mathcal{S}_4 \begin{vmatrix}
\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{100}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{100}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{100}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} \\
\frac{\alpha_{101}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{101}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{101}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} \\
\frac{\alpha_{110}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{110}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} \\
\frac{\alpha_{111}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{111}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda}
\end{vmatrix} = 0,$$

dengan $\mathcal{S}_4 = \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda}$.

9. **(OKE)**

$$\boldsymbol{K_2 \to K_2 - \frac{\alpha_{100}\alpha_{101}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2-\lambda} K_1}$$
$$\boldsymbol{K_3 \to K_3 - \frac{\alpha_{100}\alpha_{110}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2-\lambda} K_1}$$
$$\boldsymbol{K_4 \to K_4 - \frac{\alpha_{100}\alpha_{111}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2-\lambda} K_1}$$

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3\mathcal{S}_4\begin{vmatrix} \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & 0 & 0 & 0 \\ \frac{\alpha_{101}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\alpha_{101}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\alpha_{101}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} \\ \frac{\alpha_{110}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{110}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} \\ \frac{\alpha_{111}\alpha_{100}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda} & \frac{\alpha_{111}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} \end{vmatrix} = 0$$

10. **(Ekspansi baris ke-1)**

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3\mathcal{S}_4\left(\mathcal{S}_5 M_{11} - 0M_{12} + 0M_{13} - 0M_{14}\right) = 0$$

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3\mathcal{S}_4\mathcal{S}_5\begin{vmatrix} \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\alpha_{101}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\alpha_{101}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} \\ \frac{\alpha_{110}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} \\ \frac{\alpha_{111}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} \end{vmatrix} = 0,$$

dengan $\mathcal{S}_5 = \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda}$.

11. **(OKE)**
$\boldsymbol{K_2 \to K_2 - \frac{\alpha_{100}\alpha_{101}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2-\lambda}K_1}$
$\boldsymbol{K_3 \to K_3 - \frac{\alpha_{100}\alpha_{110}^*}{|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2-\lambda}K_1}$

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3\mathcal{S}_4\mathcal{S}_5\begin{vmatrix} \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & 0 & 0 \\ \frac{\alpha_{110}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2-|\alpha_{101}|^2+\lambda} & \frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} \\ \frac{\alpha_{111}\alpha_{101}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2-|\alpha_{101}|^2+\lambda} \end{vmatrix} = 0$$

12. **(Ekspansi baris ke-1)**

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3\mathcal{S}_4\mathcal{S}_5\left(\mathcal{S}_6 M_{11} - 0M_{12} + 0M_{13} - 0M_{14}\right) = 0$$

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3\mathcal{S}_4\mathcal{S}_5\mathcal{S}_6\begin{vmatrix} \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2-|\alpha_{101}|^2+\lambda} & \frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} \\ \frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda} & \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2-|\alpha_{101}|^2+\lambda} \end{vmatrix} = 0,$$

dengan $\mathcal{S}_6 = \frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda}$.

13. **(Determinan 2×2)**

$$\mathcal{S}_1\mathcal{S}_2\mathcal{S}_3\mathcal{S}_4\mathcal{S}_5\mathcal{S}_6\Bigg[\left(\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2+|\alpha_{110}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2-|\alpha_{101}|^2+\lambda}\right)$$

$$\left(\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2+|\alpha_{111}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2-|\alpha_{101}|^2+\lambda}\right)$$

$$-\left(\frac{\alpha_{110}\alpha_{111}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda}\right)$$

$$\left(\frac{\alpha_{111}\alpha_{110}^*\lambda}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda}\right)\Bigg] = 0$$

$$\left(\lambda(|\alpha_{000}|^2-\lambda)\right)\left(\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2-\lambda}{-|\alpha_{000}|^2+\lambda}\right)\left(\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2+\lambda}\right)\left(\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2+\lambda}\right)$$

$$\left(\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2+\lambda}\right)\left(\frac{\lambda(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2-\lambda)}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2+\lambda}\right)$$

$$\left[\frac{-|\alpha_{000}|^2\lambda^2-|\alpha_{001}|^2\lambda^2-|\alpha_{010}|^2\lambda^2-|\alpha_{011}|^2\lambda^2-|\alpha_{100}|^2\lambda^2-|\alpha_{101}|^2\lambda^2-|\alpha_{110}|^2\lambda^2-|\alpha_{111}|^2\lambda^2+\lambda^3}{-|\alpha_{000}|^2-|\alpha_{001}|^2-|\alpha_{010}|^2-|\alpha_{011}|^2-|\alpha_{100}|^2-|\alpha_{101}|^2+\lambda}\right] = 0$$

$$\lambda^7\left(\lambda-\left(|\alpha_{000}|^2+|\alpha_{001}|^2+|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{100}|^2+|\alpha_{101}|^2+|\alpha_{110}|^2+|\alpha_{111}|^2\right)\right) = 0.$$

Karena koefisien α_{ijk} memenuhi syarat normalisasi $\sum_{i,j,k=0}^{1}|\alpha_{ijk}|^2=1$, maka persamaan diatas menjadi

$$\lambda^7(\lambda-1)=0$$
$$\lambda_1=1 \qquad \lambda_{2,3,4,5,6,7}=0.$$

C.2.2 Keadaan Terpisah Sepenuhnya

Persamaan karakteristik matriks densitas tereduksi $\rho_A^{(sep)}$ pada persamaan (A.5) adalah

$$\left|\rho_A^{(sep)} - \lambda I\right| = 0,$$

maka

$$\begin{aligned}
\begin{vmatrix} |a_0|^2 - \lambda & a_0 a_1^* \\ a_1 a_0^* & |a_1|^2 - \lambda \end{vmatrix} &= 0 \\
|a_0|^2|a_1|^2 - \left(|a_0|^2 + |a_1|^2\right)\lambda + \lambda^2 - |a_0|^2|a_1|^2 &= 0 \\
\lambda^2 - \lambda &= 0 \\
\lambda_1 = 1 \qquad \lambda_2 &= 0.
\end{aligned} \tag{C.6}$$

Persamaan karakteristik matriks densitas tereduksi $\rho_B^{(sep)}$ pada persamaan (A.6) adalah

$$\left|\rho_B^{(sep)} - \lambda I\right| = 0,$$

maka

$$\begin{aligned}
\begin{vmatrix} |b_0|^2 - \lambda & b_0 b_1^* \\ b_1 b_0^* & |b_1|^2 - \lambda \end{vmatrix} &= 0 \\
|b_0|^2|b_1|^2 - \left(|b_0|^2 + |b_1|^2\right)\lambda + \lambda^2 - |b_0|^2|b_1|^2 &= 0 \\
\lambda^2 - \lambda &= 0 \\
\lambda_1 = 1 \qquad \lambda_2 &= 0.
\end{aligned} \tag{C.7}$$

Persamaan karakteristik matriks densitas tereduksi $\rho_C^{(sep)}$ pada persamaan (A.7) adalah

$$\left|\rho_C^{(sep)} - \lambda I\right| = 0,$$

maka

$$\begin{aligned}
\begin{vmatrix} |c_0|^2 - \lambda & c_0 c_1^* \\ c_1 c_0^* & |c_1|^2 - \lambda \end{vmatrix} &= 0 \\
|c_0|^2|c_1|^2 - \left(|c_0|^2 + |c_1|^2\right)\lambda + \lambda^2 - |c_0|^2|c_1|^2 &= 0 \\
\lambda^2 - \lambda &= 0 \\
\lambda_1 = 1 \qquad \lambda_2 &= 0.
\end{aligned} \tag{C.8}$$

Persamaan karakteristik matriks densitas tereduksi $\rho_{AB}^{(sep)}$ pada persamaan (A.8) adalah

$$\left|\rho_{AB}^{(sep)} - \lambda I\right| = 0,$$

maka

$$\begin{vmatrix}
|a_0|^2|b_0|^2 - \lambda & |a_0|^2 b_0 b_1^* & a_0 a_1^* |b_0|^2 & a_0 a_1^* b_0 b_1^* \\
|a_0|^2 b_1 b_0^* & |a_0|^2|b_1|^2 - \lambda & a_0 a_1^* b_1 b_0^* & a_0 a_1^* |b_1|^2 \\
a_1 a_0^* |b_0|^2 & a_1 a_0^* b_0 b_1^* & |a_1|^2|b_0|^2 - \lambda & |a_1|^2 b_0 b_1^* \\
a_1 a_0^* b_1 b_0^* & a_1 a_0^* |b_1|^2 & |a_1|^2 b_1 b_0^* & |a_1|^2|b_1|^2 - \lambda
\end{vmatrix} = 0 \tag{C.9}$$

Lakukan eliminasi Gauss untuk menyederhanakan determinan diatas serta melakukan ekspansi kofaktor untuk bisa menghitung nilai determinannya

1. **(OKE)**
$\boldsymbol{K_2 \to K_2 - \frac{|a_0|^2 b_0 b_1^*}{|a_0|^2|b_0|^2-\lambda} K_1}$
$\boldsymbol{K_3 \to K_3 - \frac{a_0 a_1^* |b_0|^2}{|a_0|^2|b_0|^2-\lambda} K_1}$
$\boldsymbol{K_4 \to K_4 - \frac{a_0 a_1^* b_0 b_1^*}{|a_0|^2|b_0|^2-\lambda} K_1}$

$$\begin{vmatrix} |a_0|^2|b_0|^2-\lambda & 0 & 0 & 0 \\ |a_0|^2 b_1 b_0^* & \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_0|^2|b_1|^2+\lambda\right)}{|a_0|^2|b_0|^2-\lambda} & -\frac{a_0 a_1^* b_1 b_0^* \lambda}{|a_0|^2|b_0|^2-\lambda} & -\frac{a_0 a_1^* |b_1|^2 \lambda}{|a_0|^2|b_0|^2-\lambda} \\ a_1 a_0^* |b_0|^2 & -\frac{a_1 a_0^* b_0 b_1^* \lambda}{|a_0|^2|b_0|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_1|^2|b_0|^2+\lambda\right)}{|a_0|^2|b_0|^2-\lambda} & -\frac{|a_1|^2 b_0 b_1^* \lambda}{|a_0|^2|b_0|^2-\lambda} \\ a_1 a_0^* b_1 b_0^* & -\frac{a_1 a_0^* |b_1|^2 \lambda}{|a_0|^2|b_0|^2-\lambda} & -\frac{|a_1|^2 b_1 b_0^* \lambda}{|a_0|^2|b_0|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_1|^2|b_1|^2+\lambda\right)}{|a_0|^2|b_0|^2-\lambda} \end{vmatrix} = 0$$

2. **(Ekspansi baris ke-1)**

$$(|a_0|^2|b_0|^2-\lambda)M_{11} - 0M_{12} + 0M_{13} + 0M_{14} = 0$$

$$(|a_0|^2|b_0|^2-\lambda)\begin{vmatrix} \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_0|^2|b_1|^2+\lambda\right)}{|a_0|^2|b_0|^2-\lambda} & -\frac{a_0 a_1^* b_1 b_0^* \lambda}{|a_0|^2|b_0|^2-\lambda} & -\frac{a_0 a_1^* |b_1|^2 \lambda}{|a_0|^2|b_0|^2-\lambda} \\ -\frac{a_1 a_0^* b_0 b_1^* \lambda}{|a_0|^2|b_0|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_1|^2|b_0|^2+\lambda\right)}{|a_0|^2|b_0|^2-\lambda} & -\frac{|a_1|^2 b_0 b_1^* \lambda}{|a_0|^2|b_0|^2-\lambda} \\ -\frac{a_1 a_0^* |b_1|^2 \lambda}{|a_0|^2|b_0|^2-\lambda} & -\frac{|a_1|^2 b_1 b_0^* \lambda}{|a_0|^2|b_0|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_1|^2|b_1|^2+\lambda\right)}{|a_0|^2|b_0|^2-\lambda} \end{vmatrix} = 0$$

3. **(OKE)**
$\boldsymbol{K_2 \to K_2 - \frac{a_0 a_1^* b_1 b_0^*}{|a_0|^2|b_0|^2+|a_0|^2|b_1|^2-\lambda} K_1}$
$\boldsymbol{K_3 \to K_3 - \frac{a_0 a_1^* |b_1|^2}{|a_0|^2|b_0|^2+|a_0|^2|b_1|^2-\lambda} K_1}$

$$(|a_0|^2|b_0|^2-\lambda)\begin{vmatrix} \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_0|^2|b_1|^2+\lambda\right)}{|a_0|^2|b_0|^2-\lambda} & 0 & 0 \\ -\frac{a_1 a_0^* b_0 b_1^* \lambda}{|a_0|^2|b_0|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_0|^2|b_1|^2-|a_1|^2|b_0|^2+\lambda\right)}{|a_0|^2|b_0|^2+|a_0|^2|b_1|^2-\lambda} & -\frac{|a_1|^2 b_0 b_1^* \lambda}{|a_0|^2|b_0|^2+|a_0|^2|b_1|^2-\lambda} \\ -\frac{a_1 a_0^* |b_1|^2 \lambda}{|a_0|^2|b_0|^2-\lambda} & -\frac{|a_1|^2 b_1 b_0^* \lambda}{|a_0|^2|b_0|^2+|a_0|^2|b_1|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_0|^2|b_1|^2-|a_1|^2|b_1|^2+\lambda\right)}{|a_0|^2|b_0|^2+|a_0|^2|b_1|^2-\lambda} \end{vmatrix} = 0$$

4. **(Ekspansi baris ke-1)**

$$\left(|a_0|^2|b_0|^2 - \lambda\right)\left(\frac{\lambda\left(-|a_0|^2|b_0|^2 - |a_0|^2|b_1|^2 + \lambda\right)}{|a_0|^2|b_0|^2 - \lambda}\right) M_{11} - 0M_{12} + 0M_{13} = 0$$

$$\left(|a_0|^2|b_0|^2 - \lambda\right)\left(\frac{\lambda\left(-|a_0|^2|b_0|^2 - |a_0|^2|b_1|^2 + \lambda\right)}{|a_0|^2|b_0|^2 - \lambda}\right)\begin{vmatrix} \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_0|^2|b_1|^2-|a_1|^2|b_0|^2+\lambda\right)}{|a_0|^2|b_0|^2+|a_0|^2|b_1|^2-\lambda} & -\frac{|a_1|^2 b_0 b_1^* \lambda}{|a_0|^2|b_0|^2+|a_0|^2|b_1|^2-\lambda} \\ -\frac{|a_1|^2 b_1 b_0^* \lambda}{|a_0|^2|b_0|^2+|a_0|^2|b_1|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|b_0|^2-|a_0|^2|b_1|^2-|a_1|^2|b_1|^2+\lambda\right)}{|a_0|^2|b_0|^2+|a_0|^2|b_1|^2-\lambda} \end{vmatrix} = 0$$

5. **(Determinan** 2×2**)**

$$\left(|a_0|^2|b_0|^2 - \lambda\right)\left(\frac{\lambda\left(-|a_0|^2|b_0|^2 - |a_0|^2|b_1|^2 + \lambda\right)}{|a_0|^2|b_0|^2 - \lambda}\right)\left(\frac{\lambda^2\left(|a_0|^2|b_0|^2 + |a_0|^2|b_1|^2 + |a_1|^2|b_0|^2 + |a_1|^2|b_1|^2 - \lambda\right)}{|a_0|^2|b_0|^2 + |a_0|^2|b_1|^2 - \lambda}\right) = 0$$

$$\lambda^3\left(\lambda - \left(|a_0|^2 + |a_1|^2\right)\left(|b_0|^2 + |b_1|^2\right)\right) = 0.$$

Karena koefisien a_i dan b_j masing-masing memenuhi syarat normalisasi $\sum_{i=0}^{1}|a_i|^2 = 1$ dan $\sum_{i=0}^{1}|b_i|^2 = 1$, maka persamaan diatas menjadi

$$\lambda^3(\lambda - 1) = 0$$
$$\lambda_1 = 1 \qquad \lambda_{2,3,4} = 0. \tag{C.10}$$

Persamaan karakteristik matriks densitas tereduksi $\rho_{AC}^{(sep)}$ pada persamaan (A.9) adalah

$$\left|\rho_{AC}^{(sep)} - \lambda I\right| = 0,$$

maka

$$\begin{vmatrix} |a_0|^2|c_0|^2 - \lambda & |a_0|^2 c_0 c_1^* & a_0 a_1^* |c_0|^2 & a_0 a_1^* c_0 c_1^* \\ |a_0|^2 c_1 c_0^* & |a_0|^2|c_1|^2 - \lambda & a_0 a_1^* c_1 c_0^* & a_0 a_1^* |c_1|^2 \\ a_1 a_0^* |c_0|^2 & a_1 a_0^* c_0 c_1^* & |a_1|^2|c_0|^2 - \lambda & |a_1|^2 c_0 c_1^* \\ a_1 a_0^* c_1 c_0^* & a_1 a_0^* |c_1|^2 & |a_1|^2 c_1 c_0^* & |a_1|^2|c_1|^2 - \lambda \end{vmatrix} = 0$$

(C.11)

Lakukan eliminasi Gauss untuk menyederhanakan determinan diatas serta melakukan ekspansi kofaktor untuk bisa menghitung nilai determinannya

1. **(OKE)**

   $\boldsymbol{K_2 \rightarrow K_2 - \frac{|a_0|^2 c_0 c_1^*}{|a_0|^2|c_0|^2-\lambda} K_1}$

   $\boldsymbol{K_3 \rightarrow K_3 - \frac{a_0 a_1^* |c_0|^2}{|a_0|^2|c_0|^2-\lambda} K_1}$

   $\boldsymbol{K_4 \rightarrow K_4 - \frac{a_0 a_1^* c_0 c_1^*}{|a_0|^2|c_0|^2-\lambda} K_1}$

$$\begin{vmatrix} |a_0|^2|c_0|^2-\lambda & 0 & 0 & 0 \\ |a_0|^2 c_1 c_0^* & \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_0|^2|c_1|^2+\lambda\right)}{|a_0|^2|c_0|^2-\lambda} & -\frac{a_0 a_1^* c_1 c_0^* \lambda}{|a_0|^2|c_0|^2-\lambda} & -\frac{a_0 a_1^* |c_1|^2 \lambda}{|a_0|^2|c_0|^2-\lambda} \\ a_1 a_0^* |c_0|^2 & -\frac{a_1 a_0^* c_0 c_1^* \lambda}{|a_0|^2|c_0|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_1|^2|c_0|^2+\lambda\right)}{|a_0|^2|c_0|^2-\lambda} & -\frac{|a_1|^2 c_0 c_1^* \lambda}{|a_0|^2|c_0|^2-\lambda} \\ a_1 a_0^* c_1 c_0^* & -\frac{a_1 a_0^* |c_1|^2 \lambda}{|a_0|^2|c_0|^2-\lambda} & -\frac{|a_1|^2 c_1 c_0^* \lambda}{|a_0|^2|c_0|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_1|^2|c_1|^2+\lambda\right)}{|a_0|^2|c_0|^2-\lambda} \end{vmatrix} = 0$$

2. **(Ekspansi baris ke-1)**

$$(|a_0|^2|c_0|^2-\lambda)M_{11} - 0M_{12} + 0M_{13} + 0M_{14} = 0$$

$$(|a_0|^2|c_0|^2-\lambda)\begin{vmatrix} \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_0|^2|c_1|^2+\lambda\right)}{|a_0|^2|c_0|^2-\lambda} & -\frac{a_0 a_1^* c_1 c_0^* \lambda}{|a_0|^2|c_0|^2-\lambda} & -\frac{a_0 a_1^* |c_1|^2 \lambda}{|a_0|^2|c_0|^2-\lambda} \\ -\frac{a_1 a_0^* c_0 c_1^* \lambda}{|a_0|^2|c_0|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_1|^2|c_0|^2+\lambda\right)}{|a_0|^2|c_0|^2-\lambda} & -\frac{|a_1|^2 c_0 c_1^* \lambda}{|a_0|^2|c_0|^2-\lambda} \\ -\frac{a_1 a_0^* |c_1|^2 \lambda}{|a_0|^2|c_0|^2-\lambda} & -\frac{|a_1|^2 c_1 c_0^* \lambda}{|a_0|^2|c_0|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_1|^2|c_1|^2+\lambda\right)}{|a_0|^2|c_0|^2-\lambda} \end{vmatrix} = 0$$

3. **(OKE)**

   $\boldsymbol{K_2 \rightarrow K_2 - \frac{a_0 a_1^* c_1 c_0^*}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda} K_1}$

   $\boldsymbol{K_3 \rightarrow K_3 - \frac{a_0 a_1^* |c_1|^2}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda} K_1}$

$$(|a_0|^2|c_0|^2-\lambda)\begin{vmatrix} \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_0|^2|c_1|^2+\lambda\right)}{|a_0|^2|c_0|^2-\lambda} & 0 & 0 \\ -\frac{a_1a_0^*c_0c_1^*\lambda}{|a_0|^2|c_0|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_0|^2|c_1|^2-|a_1|^2|c_0|^2+\lambda\right)}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda} & -\frac{|a_1|^2c_0c_1^*\lambda}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda} \\ -\frac{a_1a_0^*|c_1|^2\lambda}{|a_0|^2|c_0|^2-\lambda} & -\frac{|a_1|^2c_1c_0^*\lambda}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_0|^2|c_1|^2-|a_1|^2|c_1|^2+\lambda\right)}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda} \end{vmatrix}=0$$

4. **(Ekspansi baris ke-1)**

$$(|a_0|^2|c_0|^2-\lambda)\left(\frac{\lambda\left(-|a_0|^2|c_0|^2-|a_0|^2|c_1|^2+\lambda\right)}{|a_0|^2|c_0|^2-\lambda}\right)M_{11}-0M_{12}+0M_{13}=0$$

$$(|a_0|^2|c_0|^2-\lambda)\left(\frac{\lambda\left(-|a_0|^2|c_0|^2-|a_0|^2|c_1|^2+\lambda\right)}{|a_0|^2|c_0|^2-\lambda}\right)\begin{vmatrix} \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_0|^2|c_1|^2-|a_1|^2|c_0|^2+\lambda\right)}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda} & -\frac{|a_1|^2c_0c_1^*\lambda}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda} \\ -\frac{|a_1|^2c_1c_0^*\lambda}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda} & \frac{\lambda\left(-|a_0|^2|c_0|^2-|a_0|^2|c_1|^2-|a_1|^2|c_1|^2+\lambda\right)}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda} \end{vmatrix}=0$$

5. **(Determinan 2×2)**

$$(|a_0|^2|c_0|^2-\lambda)\left(\frac{\lambda\left(-|a_0|^2|c_0|^2-|a_0|^2|c_1|^2+\lambda\right)}{|a_0|^2|c_0|^2-\lambda}\right)\left(\frac{\lambda^2\left(|a_0|^2|c_0|^2+|a_0|^2|c_1|^2+|a_1|^2|c_0|^2+|a_1|^2|c_1|^2-\lambda\right)}{|a_0|^2|c_0|^2+|a_0|^2|c_1|^2-\lambda}\right)=0$$

$$\lambda^3\left(\lambda-\left(|a_0|^2+|a_1|^2\right)\left(|c_0|^2+|c_1|^2\right)\right)=0.$$

Karena koefisien a_i dan c_j masing-masing memenuhi syarat normalisasi $\sum_{i=0}^{1}|a_i|^2=1$ dan $\sum_{i=0}^{1}|c_i|^2=1$, maka persamaan diatas menjadi

$$\begin{aligned}\lambda^3(\lambda-1)&=0\\ \lambda_1=1 \qquad \lambda_{2,3,4}&=0.\end{aligned} \tag{C.12}$$

Persamaan karakteristik matriks densitas tereduksi $\rho_{BC}^{(sep)}$ pada persamaan (A.9) adalah

$$\left|\rho_{BC}^{(sep)}-\lambda I\right|=0,$$

maka

$$\begin{vmatrix} |b_0|^2|c_0|^2-\lambda & |b_0|^2c_0c_1^* & b_0b_1^*|c_0|^2 & b_0b_1^*c_0c_1^* \\ |b_0|^2c_1c_0^* & |b_0|^2|c_1|^2-\lambda & b_0b_1^*c_1c_0^* & b_0b_1^*|c_1|^2 \\ b_1b_0^*|c_0|^2 & b_1b_0^*c_0c_1^* & |b_1|^2|c_0|^2-\lambda & |b_1|^2c_0c_1^* \\ b_1b_0^*c_1c_0^* & b_1b_0^*|c_1|^2 & |b_1|^2c_1c_0^* & |b_1|^2|c_1|^2-\lambda \end{vmatrix} = 0 \tag{C.13}$$

Lakukan eliminasi Gauss untuk menyederhanakan determinan diatas serta melakukan ekspansi kofaktor untuk bisa menghitung nilai determinannya

1. **(OKE)**
   $\boldsymbol{K_2 \to K_2 - \frac{|b_0|^2c_0c_1^*}{|b_0|^2|c_0|^2-\lambda}K_1}$
   $\boldsymbol{K_3 \to K_3 - \frac{b_0b_1^*|c_0|^2}{|b_0|^2|c_0|^2-\lambda}K_1}$
   $\boldsymbol{K_4 \to K_4 - \frac{b_0b_1^*c_0c_1^*}{|b_0|^2|c_0|^2-\lambda}K_1}$

$$\begin{vmatrix} |b_0|^2|c_0|^2-\lambda & 0 & 0 & 0 \\ |b_0|^2c_1c_0^* & \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_0|^2|c_1|^2+\lambda\right)}{|b_0|^2|c_0|^2-\lambda} & -\frac{b_0b_1^*c_1c_0^*\lambda}{|b_0|^2|c_0|^2-\lambda} & -\frac{b_0b_1^*|c_1|^2\lambda}{|b_0|^2|c_0|^2-\lambda} \\ b_1b_0^*|c_0|^2 & -\frac{b_1b_0^*c_0c_1^*\lambda}{|b_0|^2|c_0|^2-\lambda} & \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_1|^2|c_0|^2+\lambda\right)}{|b_0|^2|c_0|^2-\lambda} & -\frac{|b_1|^2c_0c_1^*\lambda}{|b_0|^2|c_0|^2-\lambda} \\ b_1b_0^*c_1c_0^* & -\frac{b_1b_0^*|c_1|^2\lambda}{|b_0|^2|c_0|^2-\lambda} & -\frac{|b_1|^2c_1c_0^*\lambda}{|b_0|^2|c_0|^2-\lambda} & \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_1|^2|c_1|^2+\lambda\right)}{|b_0|^2|c_0|^2-\lambda} \end{vmatrix} = 0$$

2. **(Ekspansi baris ke-1)**

$$(|b_0|^2|c_0|^2-\lambda)M_{11} - 0M_{12} + 0M_{13} + 0M_{14} = 0$$

$$(|b_0|^2|c_0|^2-\lambda)\begin{vmatrix} \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_0|^2|c_1|^2+\lambda\right)}{|b_0|^2|c_0|^2-\lambda} & -\frac{b_0b_1^*c_1c_0^*\lambda}{|b_0|^2|c_0|^2-\lambda} & -\frac{b_0b_1^*|c_1|^2\lambda}{|b_0|^2|c_0|^2-\lambda} \\ -\frac{b_1b_0^*c_0c_1^*\lambda}{|b_0|^2|c_0|^2-\lambda} & \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_1|^2|c_0|^2+\lambda\right)}{|b_0|^2|c_0|^2-\lambda} & -\frac{|b_1|^2c_0c_1^*\lambda}{|b_0|^2|c_0|^2-\lambda} \\ -\frac{b_1b_0^*|c_1|^2\lambda}{|b_0|^2|c_0|^2-\lambda} & -\frac{|b_1|^2c_1c_0^*\lambda}{|b_0|^2|c_0|^2-\lambda} & \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_1|^2|c_1|^2+\lambda\right)}{|b_0|^2|c_0|^2-\lambda} \end{vmatrix} = 0$$

3. **(OKE)**
$\boldsymbol{K_2 \to K_2 - \frac{b_0 b_1^* c_1 c_0^*}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda} K_1}$
$\boldsymbol{K_3 \to K_3 - \frac{b_0 b_1^* |c_1|^2}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda} K_1}$

$$(|b_0|^2|c_0|^2-\lambda)\begin{vmatrix} \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_0|^2|c_1|^2+\lambda\right)}{|b_0|^2|c_0|^2-\lambda} & 0 & 0 \\ -\frac{b_1 b_0^* c_0 c_1^* \lambda}{|b_0|^2|c_0|^2-\lambda} & \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_0|^2|c_1|^2-|b_1|^2|c_0|^2+\lambda\right)}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda} & -\frac{|b_1|^2 c_0 c_1^* \lambda}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda} \\ -\frac{b_1 b_0^* |c_1|^2 \lambda}{|b_0|^2|c_0|^2-\lambda} & -\frac{|b_1|^2 c_1 c_0^* \lambda}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda} & \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_0|^2|c_1|^2-|b_1|^2|c_1|^2+\lambda\right)}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda} \end{vmatrix} = 0$$

4. **(Ekspansi baris ke-1)**

$$(|b_0|^2|c_0|^2-\lambda)\left(\frac{\lambda\left(-|b_0|^2|c_0|^2-|b_0|^2|c_1|^2+\lambda\right)}{|b_0|^2|c_0|^2-\lambda}\right) M_{11} - 0M_{12} + 0M_{13} = 0$$

$$(|b_0|^2|c_0|^2-\lambda)\left(\frac{\lambda\left(-|b_0|^2|c_0|^2-|b_0|^2|c_1|^2+\lambda\right)}{|b_0|^2|c_0|^2-\lambda}\right)\begin{vmatrix} \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_0|^2|c_1|^2-|b_1|^2|c_0|^2+\lambda\right)}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda} & -\frac{|b_1|^2 c_0 c_1^* \lambda}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda} \\ -\frac{|b_1|^2 c_1 c_0^* \lambda}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda} & \frac{\lambda\left(-|b_0|^2|c_0|^2-|b_0|^2|c_1|^2-|b_1|^2|c_1|^2+\lambda\right)}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda} \end{vmatrix} = 0$$

5. **(Determinan 2×2)**

$$(|b_0|^2|c_0|^2-\lambda)\left(\frac{\lambda\left(-|b_0|^2|c_0|^2-|b_0|^2|c_1|^2+\lambda\right)}{|b_0|^2|c_0|^2-\lambda}\right)\left(\frac{\lambda^2\left(|b_0|^2|c_0|^2+|b_0|^2|c_1|^2+|b_1|^2|c_0|^2+|b_1|^2|c_1|^2-\lambda\right)}{|b_0|^2|c_0|^2+|b_0|^2|c_1|^2-\lambda}\right) = 0$$
$$\lambda^3\left(\lambda - \left(|b_0|^2+|b_1|^2\right)\left(|c_0|^2+|c_1|^2\right)\right) = 0.$$

Karena koefisien b_i dan c_j masing-masing memenuhi syarat normalisasi $\sum_{i=0}^{1}|b_i|^2 = 1$ dan $\sum_{i=0}^{1}|c_i|^2 = 1$, maka persamaan diatas menjadi

$$\lambda^3(\lambda-1) = 0$$
$$\lambda_1 = 1 \qquad \lambda_{2,3,4} = 0. \tag{C.14}$$

C.2.3 Keadaan Terbelit Sepenuhnya

Persamaan karakteristik matriks densitas tereduksi $\rho_A^{(ent)}$ pada persamaan (A.11) adalah

$$\left|\rho_A^{(ent)} - \lambda I\right| = 0,$$

maka

$$\begin{vmatrix} |\alpha_{000}|^2 + |\alpha_{001}|^2 + |\alpha_{010}|^2 + |\alpha_{011}|^2 - \lambda & \alpha_{000}\alpha_{100}^* + \alpha_{001}\alpha_{101}^* + \alpha_{010}\alpha_{110}^* + \alpha_{011}\alpha_{111}^* \\ \alpha_{100}\alpha_{000}^* + \alpha_{101}\alpha_{001}^* + \alpha_{110}\alpha_{010}^* + \alpha_{111}\alpha_{011}^* & |\alpha_{100}|^2 + |\alpha_{101}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2 - \lambda \end{vmatrix} = 0$$

$$\left(|\alpha_{000}|^2 + |\alpha_{001}|^2 + |\alpha_{010}|^2 + |\alpha_{011}|^2 - \lambda\right)\left(|\alpha_{100}|^2 + |\alpha_{101}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2 - \lambda\right)$$
$$- \left(\alpha_{000}\alpha_{100}^* + \alpha_{001}\alpha_{101}^* + \alpha_{010}\alpha_{110}^* + \alpha_{011}\alpha_{111}^*\right)\left(\alpha_{100}\alpha_{000}^* + \alpha_{101}\alpha_{001}^* + \alpha_{110}\alpha_{010}^* + \alpha_{111}\alpha_{011}^*\right) = 0$$
$$\begin{aligned} \lambda^2 - \lambda + \Big(& |\alpha_{000}|^2\left(|\alpha_{101}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2\right) + |\alpha_{001}|^2\left(|\alpha_{100}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2\right) \\ & |\alpha_{010}|^2\left(|\alpha_{100}|^2 + |\alpha_{101}|^2 + |\alpha_{111}|^2\right) + |\alpha_{011}|^2\left(|\alpha_{100}|^2 + |\alpha_{101}|^2 + |\alpha_{110}|^2\right) \end{aligned}$$
$$-\alpha_{000}\alpha_{100}^*\left(\alpha_{101}\alpha_{001}^* + \alpha_{110}\alpha_{010}^* + \alpha_{111}\alpha_{011}^*\right) - \alpha_{001}\alpha_{101}^*\left(\alpha_{000}\alpha_{100}^* + \alpha_{010}\alpha_{110}^* + \alpha_{011}\alpha_{111}^*\right)$$
$$-\alpha_{010}\alpha_{110}^*\left(\alpha_{100}\alpha_{000}^* + \alpha_{101}\alpha_{001}^* + \alpha_{111}\alpha_{011}^*\right) - \alpha_{011}\alpha_{111}^*\left(\alpha_{000}\alpha_{100}^* + \alpha_{001}\alpha_{101}^* + \alpha_{010}\alpha_{110}^*\right)\Big) = 0$$

$$\lambda_{1,2} = \frac{1}{2}\left(1 \pm \sqrt{D_A^{(ent,3)}}\right), \tag{C.15}$$

dimana

$$\begin{aligned} D_A^{(ent,3)} = 1 - 4\Big(& |\alpha_{000}|^2\left(|\alpha_{101}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2\right) + |\alpha_{001}|^2\left(|\alpha_{100}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2\right) \\ & + |\alpha_{010}|^2\left(|\alpha_{100}|^2 + |\alpha_{101}|^2 + |\alpha_{111}|^2\right) + |\alpha_{011}|^2\left(|\alpha_{100}|^2 + |\alpha_{101}|^2 + |\alpha_{110}|^2\right) \\ & - \alpha_{000}\alpha_{100}^*\left(\alpha_{101}\alpha_{001}^* + \alpha_{110}\alpha_{010}^* + \alpha_{111}\alpha_{011}^*\right) \\ & - \alpha_{001}\alpha_{101}^*\left(\alpha_{100}\alpha_{000}^* + \alpha_{110}\alpha_{010}^* + \alpha_{111}\alpha_{011}^*\right) \\ & - \alpha_{010}\alpha_{110}^*\left(\alpha_{100}\alpha_{000}^* + \alpha_{101}\alpha_{001}^* + \alpha_{111}\alpha_{011}^*\right) \\ & - \alpha_{011}\alpha_{111}^*\left(\alpha_{100}\alpha_{000}^* + \alpha_{101}\alpha_{001}^* + \alpha_{110}\alpha_{010}^*\right)\Big) \end{aligned}$$

Persamaan karakteristik matriks densitas tereduksi $\rho_B^{(ent)}$ pada persamaan (A.12) adalah

$$\left|\rho_B^{(ent)} - \lambda I\right| = 0,$$

maka

$$\begin{vmatrix} |\alpha_{000}|^2 + |\alpha_{001}|^2 + |\alpha_{100}|^2 + |\alpha_{101}|^2 - \lambda & \alpha_{000}\alpha_{010}^* + \alpha_{001}\alpha_{011}^* + \alpha_{100}\alpha_{110}^* + \alpha_{101}\alpha_{111}^* \\ \alpha_{010}\alpha_{000}^* + \alpha_{011}\alpha_{001}^* + \alpha_{110}\alpha_{100}^* + \alpha_{111}\alpha_{101}^* & |\alpha_{010}|^2 + |\alpha_{011}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2 - \lambda \end{vmatrix} = 0$$

$$\left(|\alpha_{000}|^2 + |\alpha_{001}|^2 + |\alpha_{100}|^2 + |\alpha_{101}|^2 - \lambda\right)\left(|\alpha_{010}|^2 + |\alpha_{011}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2 - \lambda\right)$$
$$- \left(\alpha_{000}\alpha_{010}^* + \alpha_{001}\alpha_{011}^* + \alpha_{100}\alpha_{110}^* + \alpha_{101}\alpha_{111}^*\right)\left(\alpha_{010}\alpha_{000}^* + \alpha_{011}\alpha_{001}^* + \alpha_{110}\alpha_{100}^* + \alpha_{111}\alpha_{101}^*\right) = 0$$
$$\begin{aligned} \lambda^2 - \lambda + \Big(& |\alpha_{000}|^2\left(|\alpha_{011}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2\right) + |\alpha_{001}|^2\left(|\alpha_{010}|^2 + |\alpha_{110}|^2 + |\alpha_{111}|^2\right) \\ & |\alpha_{100}|^2\left(|\alpha_{010}|^2 + |\alpha_{011}|^2 + |\alpha_{111}|^2\right) + |\alpha_{101}|^2\left(|\alpha_{010}|^2 + |\alpha_{011}|^2 + |\alpha_{110}|^2\right) \end{aligned}$$

$$-\alpha_{000}\alpha_{010}^{*}\left(\alpha_{011}\alpha_{001}^{*}+\alpha_{110}\alpha_{100}^{*}+\alpha_{111}\alpha_{101}^{*}\right)-\alpha_{001}\alpha_{011}^{*}\left(\alpha_{010}\alpha_{000}^{*}+\alpha_{110}\alpha_{100}^{*}+\alpha_{111}\alpha_{101}^{*}\right)$$
$$-\alpha_{100}\alpha_{110}^{*}\left(\alpha_{010}\alpha_{000}^{*}+\alpha_{011}\alpha_{001}^{*}+\alpha_{111}\alpha_{101}^{*}\right)-\alpha_{101}\alpha_{111}^{*}\left(\alpha_{010}\alpha_{000}^{*}+\alpha_{011}\alpha_{001}^{*}+\alpha_{110}\alpha_{100}^{*}\right)\Bigg)=0$$

$$\lambda_{1,2}=\frac{1}{2}\left(1\pm\sqrt{D_B^{(ent,3)}}\right), \tag{C.16}$$

dimana

$$\begin{aligned}D_B^{(ent,3)}=1-4\Bigg(&|\alpha_{000}|^2\left(|\alpha_{011}|^2+|\alpha_{110}|^2+|\alpha_{111}|^2\right)+|\alpha_{001}|^2\left(|\alpha_{010}|^2+|\alpha_{110}|^2+|\alpha_{111}|^2\right)\\&|\alpha_{100}|^2\left(|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{111}|^2\right)+|\alpha_{101}|^2\left(|\alpha_{010}|^2+|\alpha_{011}|^2+|\alpha_{110}|^2\right)\\&-\alpha_{000}\alpha_{010}^{*}\left(\alpha_{011}\alpha_{001}^{*}+\alpha_{110}\alpha_{100}^{*}+\alpha_{111}\alpha_{101}^{*}\right)\\&-\alpha_{001}\alpha_{011}^{*}\left(\alpha_{010}\alpha_{000}^{*}+\alpha_{110}\alpha_{100}^{*}+\alpha_{111}\alpha_{101}^{*}\right)\\&-\alpha_{100}\alpha_{110}^{*}\left(\alpha_{010}\alpha_{000}^{*}+\alpha_{011}\alpha_{001}^{*}+\alpha_{111}\alpha_{101}^{*}\right)\\&-\alpha_{101}\alpha_{111}^{*}\left(\alpha_{010}\alpha_{000}^{*}+\alpha_{011}\alpha_{001}^{*}+\alpha_{110}\alpha_{100}^{*}\right)\Bigg)\end{aligned}$$

Persamaan karakteristik matriks densitas tereduksi $\rho_C^{(ent)}$ pada persamaan (A.13) adalah

$$\left|\rho_C^{(ent)}-\lambda I\right|=0,$$

maka

$$\begin{vmatrix}|\alpha_{000}|^2+|\alpha_{010}|^2+|\alpha_{100}|^2+|\alpha_{110}|^2-\lambda & \alpha_{000}\alpha_{001}^{*}+\alpha_{010}\alpha_{011}^{*}+\alpha_{100}\alpha_{101}^{*}+\alpha_{110}\alpha_{111}^{*}\\ \alpha_{001}\alpha_{000}^{*}+\alpha_{011}\alpha_{010}^{*}+\alpha_{101}\alpha_{100}^{*}+\alpha_{111}\alpha_{110}^{*} & |\alpha_{001}|^2+|\alpha_{011}|^2+|\alpha_{101}|^2+|\alpha_{111}|^2-\lambda\end{vmatrix}=0$$
$$\left(|\alpha_{000}|^2+|\alpha_{010}|^2+|\alpha_{100}|^2+|\alpha_{110}|^2-\lambda\right)\left(|\alpha_{001}|^2+|\alpha_{011}|^2+|\alpha_{101}|^2+|\alpha_{111}|^2-\lambda\right)$$
$$-\left(\alpha_{000}\alpha_{001}^{*}+\alpha_{010}\alpha_{011}^{*}+\alpha_{100}\alpha_{101}^{*}+\alpha_{110}\alpha_{111}^{*}\right)\left(\alpha_{001}\alpha_{000}^{*}+\alpha_{011}\alpha_{010}^{*}+\alpha_{101}\alpha_{100}^{*}+\alpha_{111}\alpha_{110}^{*}\right)=0$$
$$\lambda^2-\lambda+\Bigg(|\alpha_{000}|^2\left(|\alpha_{011}|^2+|\alpha_{101}|^2+|\alpha_{111}|^2\right)+|\alpha_{100}|^2\left(|\alpha_{001}|^2+|\alpha_{101}|^2+|\alpha_{111}|^2\right)$$
$$|\alpha_{100}|^2\left(|\alpha_{001}|^2+|\alpha_{011}|^2+|\alpha_{111}|^2\right)+|\alpha_{110}|^2\left(|\alpha_{001}|^2+|\alpha_{011}|^2+|\alpha_{101}|^2\right)$$
$$-\alpha_{000}\alpha_{010}^{*}\left(\alpha_{011}\alpha_{010}^{*}+\alpha_{101}\alpha_{100}^{*}+\alpha_{111}\alpha_{110}^{*}\right)-\alpha_{001}\alpha_{011}^{*}\left(\alpha_{001}\alpha_{000}^{*}+\alpha_{101}\alpha_{100}^{*}+\alpha_{111}\alpha_{110}^{*}\right)$$
$$-\alpha_{100}\alpha_{110}^{*}\left(\alpha_{001}\alpha_{000}^{*}+\alpha_{011}\alpha_{010}^{*}+\alpha_{111}\alpha_{110}^{*}\right)-\alpha_{110}\alpha_{111}^{*}\left(\alpha_{010}\alpha_{000}^{*}+\alpha_{011}\alpha_{001}^{*}+\alpha_{110}\alpha_{100}^{*}\right)\Bigg)=0$$

$$\lambda_{1,2}=\frac{1}{2}\left(1\pm\sqrt{D_C^{(ent,3)}}\right), \tag{C.17}$$

dimana

$$\begin{aligned}D_C^{(ent,3)}=1-4\Bigg(&|\alpha_{000}|^2\left(|\alpha_{011}|^2+|\alpha_{101}|^2+|\alpha_{111}|^2\right)+|\alpha_{100}|^2\left(|\alpha_{001}|^2+|\alpha_{101}|^2+|\alpha_{111}|^2\right)\\&|\alpha_{100}|^2\left(|\alpha_{001}|^2+|\alpha_{011}|^2+|\alpha_{111}|^2\right)+|\alpha_{110}|^2\left(|\alpha_{001}|^2+|\alpha_{011}|^2+|\alpha_{101}|^2\right)\\&-\alpha_{000}\alpha_{010}^{*}\left(\alpha_{011}\alpha_{010}^{*}+\alpha_{101}\alpha_{100}^{*}+\alpha_{111}\alpha_{110}^{*}\right)\\&-\alpha_{001}\alpha_{011}^{*}\left(\alpha_{001}\alpha_{000}^{*}+\alpha_{101}\alpha_{100}^{*}+\alpha_{111}\alpha_{110}^{*}\right)\\&-\alpha_{100}\alpha_{110}^{*}\left(\alpha_{001}\alpha_{000}^{*}+\alpha_{011}\alpha_{010}^{*}+\alpha_{111}\alpha_{110}^{*}\right)\\&-\alpha_{110}\alpha_{111}^{*}\left(\alpha_{010}\alpha_{000}^{*}+\alpha_{011}\alpha_{001}^{*}+\alpha_{110}\alpha_{100}^{*}\right)\Bigg)\end{aligned}$$

Halaman ini sengaja dikosongkan